Group 12 — Multimodal AI Final Project Report

Continuous Assessment 4 · Consolidated project report (CA1–CA4)

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This report consolidates the problem definition and literature review from Report 1, resources/dataset/EDA from Report 2, first implementation results from CA3, and the final implementation, experiments and fusion comparison for CA4. It is one group report. Earlier submitted PDFs remain unchanged as supporting records.

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One cumulative group report. The supporting record PDF appends earlier submissions unchanged.

Abstract

InfluenceIQ is an offline meeting-analysis prototype that connects speaker-attributed discussion to presentation content and exposes candidate actions, decisions and participant contributions with source references. The project combines pretrained speech recognition, speaker diarization, native PPT extraction and deterministic NLP analysis. The laptop implementation uses faster-whisper, local sherpa-onnx speaker models, React, FastAPI and SQLite; the Android implementation uses whisper.cpp, packaged speaker models and native Java analysis. Generative meeting notes and question answering are excluded from the final analytical workflow.

The final assessment compares three project alignment configurations: temporal-only, semantic-only and combined temporal–semantic alignment. A reproducible diagnostic uses 24 author-designed cases and six constructed slides, with the pretrained sentence encoder frozen. Exact top-1 correctness was 12/24 (50.00%), 20/24 (83.33%) and 16/24 (66.67%), respectively. These results describe the controlled fixture, not real-meeting alignment accuracy. On a separate saved real recording with 1,008 transcript passages and 34 untimed slides, semantic and combined methods both linked 39 passages; independent relevance labels were unavailable. This demonstrates the implemented missing-timing fallback rather than a measured accuracy improvement.

Existing annotated English AMI experiments provide separately scoped speech results: laptop small-model WER 42.30% on a ten-minute window; Android base-model WER 37.74% and native speaker DER 13.48% on a different 90-second window. The windows, models and execution paths differ, so these are not a direct platform comparison. The project delivers a usable local workflow and inspectable analysis, while natural Hindi/Hinglish accuracy, genuine timed-slide evaluation, human influence agreement and physical-phone validation remain research work.

Keywords: multimodal meeting intelligence; offline inference; speaker diarization; transcript–slide alignment; temporal fusion; semantic fusion; decision lineage; participant influence; extractive NLP.

1. Assessment coverage and evidence policy

The final-assessment instruction asks for results for the fusion techniques employed, a comparison of all three techniques, and a single report carrying the group number, member names and PRNs. Those details appear on the first page. The cumulative coverage is:

Assessment / source	Contribution incorporated in this report
Report 1, 29 pages	Problem, motivation, objectives, literature themes, planned architecture and evaluation
Report 2, 11 pages	Resource allocation, risks, AMI dataset, selected modalities and EDA
CA3, 10 pages	First pipeline output, decision/action examples and influence-versus-participation observations
Individual literature review, 24 pages	Speech robustness, overlapping speech, speaker attribution and research protocol
CA4 / current repository	Final delivered application, actual fusion implementation, reproducible comparisons, measured speech/performance results and limitations

The fusion comparison follows the project's implemented audio–presentation alignment: **temporal alignment, semantic alignment and their combined temporal–semantic scoring**. The final assessment isolates the first two as temporal-only and semantic-only baselines and compares them with the combined aligner. Section 6 maps each technique to the actual code and explains how it connects the two input modalities. Early/intermediate/late terminology is background; it is not a claim that three separately trained neural architectures were built.

Results are classified as: **annotated real-data measurements**, **controlled diagnostic results**, **unlabelled real-data outputs**, **historical application outputs**, and **engineering checks**. A count of extracted decisions, a model score or a passing software test is not an accuracy measurement. Accuracy needs a suitable reference annotation. All new fixture labels are explicitly author-designed; no independent human evaluation or training is asserted.

2. Introduction, problem and objectives

Meetings contain both spoken deliberation and documents that provide context, alternatives and evidence. A transcript preserves words but does not automatically explain which proposal was revised, which objection was addressed, which slide was relevant or whose contribution influenced a candidate decision. Manual review becomes costly for long recordings and multiple presentations.

The project addresses an engineering and research question: **can locally processed audio and presentation content be connected to an inspectable chain of discussion events and participant-contribution estimates?** It uses speaking participation as a descriptive baseline while modelling decision-linked evidence and interactions separately. A score is an analytical estimate, not proof of causal influence or organizational authority.

The objectives, carried forward from Report 1, are:

1. Accept meeting recordings and produce timestamped, speaker-attributed transcripts.
2. Extract slide titles, body text, tables and notes from presentations.
3. Align discussion passages with slides using content and, where available, display-time information.
4. Extract candidate proposals, objections, evidence, revisions, agreements, decisions and actions.
5. Preserve source paths through evidence relations, candidate decision lineage and participant interactions.
6. Compare alignment configurations and report their measured behavior and failure cases.
7. Deliver local review interfaces with editable speaker names/roles, playback, exports and persistent history.

The project prioritizes local processing and reviewability. Inference can work without connectivity after model provisioning. Optional remote workstation processing is a separate deployment mode requiring access to that server; it does not make remote processing offline from the client’s perspective.

3. Literature review and research positioning

The previous submissions reviewed speech enhancement, multi-talker recognition, diarization, multimodal alignment and meeting roles/influence. They motivate a modular pipeline in which errors and timestamps remain visible across stages. The final report retains that foundation without implying that every reviewed model has been implemented.

Literature theme	Relevance to InfluenceIQ	Final implementation boundary
Speech enhancement / separation	Noise, reverberation and overlapping speech affect perception	No newly trained separation or enhancement network is deployed
Speaker diarization / attributed ASR	Links words to speaker-time intervals	Pretrained local inference and overlap-based attribution are implemented
Multimodal representation / fusion	Distinguishes feature, learned-representation and decision fusion	Current slide matching combines separate content/time scores [2]
Sentence embeddings	Supports paraphrase-sensitive text relevance	Frozen all-MiniLM-L6-v2 is used for Python semantic matching [3], [6]
Speech/video-to-slide alignment	Shows the value of document and visual evidence	Native slide text is implemented; visual encoders/OCR are not [4]
Participant roles / dominance	Participation and decision contribution are different constructs	Heuristic contribution scoring exists; human agreement remains unmeasured

MaViLS is relevant prior work on speech, OCR and visual features for video-to-slide alignment [4]. Its published benchmark numbers are not results of this project. The supplied individual review discusses low-latency and overlap approaches, but they remain enabling literature rather than automatically implemented features.

The project's contribution is an **offline integration with source-linked decision and contribution analysis**, not the invention of Whisper, sentence embeddings or diarization. Evidence paths can help users review estimates, but extractive output is not guaranteed correct: ASR errors, speaker confusion, keyword false positives and incorrect slide grounding can propagate. A claim of research superiority would require matched labelled meetings and independently assessed influence; that experiment has not been completed.

4. Dataset, modalities and corrected EDA

4.1 Dataset and selected modalities

The AMI Meeting Corpus provides approximately 100 hours of synchronized meeting recordings, including scenario-based design-team meetings and natural meetings, with transcripts and other annotations [1]. This project uses meeting audio and extracted presentation/document content. Video exists in the corpus but is not part of the deployed fusion pipeline.

Audio supplies what was said, when and by which inferred speaker. Presentation text supplies topic, alternatives, quantities and document context. Acoustic audio reaches the fusion stage through ASR text/timestamps; the slide branch uses extracted text, tables and notes. Images are not converted to learned visual representations. The system therefore performs **audio-derived text to document-text alignment**, not an end-to-end acoustic–image network.

A local inventory on 10 October found 2,039 AMI WAV files occupying approximately 138.03 GB (128.55 GiB), 171 WAV inputs under `data/raw/audio/`, and 987 PPTX files under `data/raw/ppt/`. These are file counts, including channels/document variants; they are not counts of independent labelled meetings or complete paired experiments. The local storage total need not equal the earlier estimate of more than 80 GB because acquisition coverage and channels differ.

4.2 Dialogue-act counts

A new XML enumeration of `ami_public_manual_1.6.2/dialogueActs/*.dialog-act.xml` used 556 files and the corpus's `ontologies/da-types.xml` labels. It counted **117,887 dialogue acts**:

Ontology category	Count	Share
Inform	33,484	28.40%
Assess	21,391	18.15%
Fragment	16,796	14.25%
Backchannel	12,573	10.67%
Suggest	9,391	7.97%
Stall	8,459	7.18%
Elicit-Inform	4,240	3.60%
Comment-About-Understanding	2,367	2.01%
Elicit-Assessment	2,265	1.92%
Be-Positive	2,210	1.87%
Other	2,204	1.87%
Offer	1,530	1.30%
Elicit-Offer-Or-Suggestion	686	0.58%

Ontology category	Count	Share
Elicit-Comment-Understanding	193	0.16%
Be-Negative	98	0.08%
Total	**117,887**	**100% before rounding**

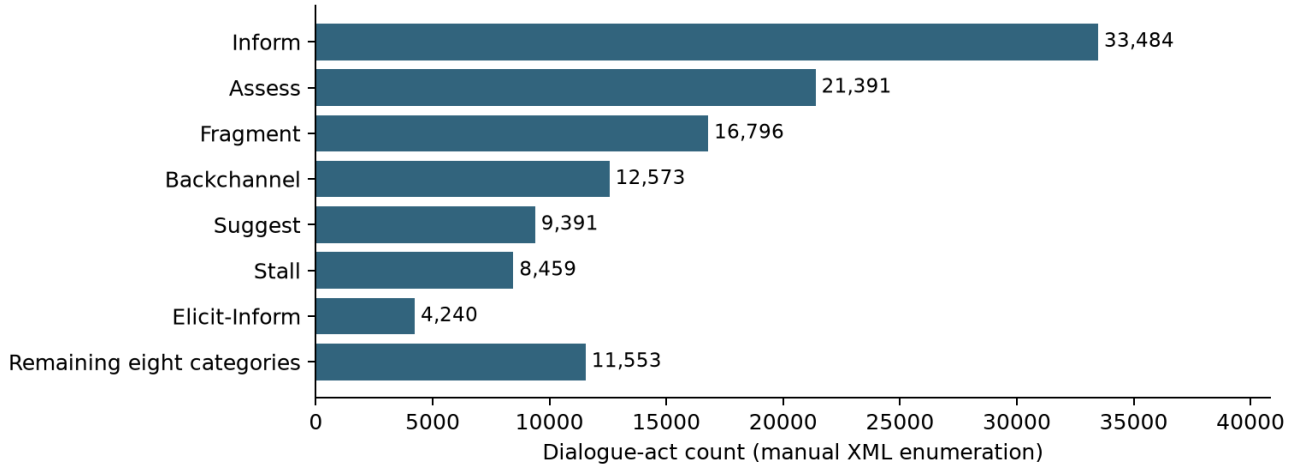


Figure 1. Re-enumerated local manual dialogue acts. Small categories are grouped only in this chart; the table preserves all ontology labels.

Report 2's total of 116,357 is 1,530 lower, exactly the Offer count. Its Backchannel value of 21,910 also pools several other categories rather than representing the literal Backchannel label, while 686 corresponds to Elicit-Offer-Or-Suggestion. The final table uses the ontology directly and supersedes that earlier grouped summary. Dialogue-act counts describe the corpus; they are not event-extraction performance or influence labels.

4.3 Preprocessing implications

Variable sample rates/channels require normalization to 16 kHz mono PCM. Noisy or overlapping speech requires cautious speaker attribution; an ambiguous speaker can remain `UNKNOWN`. Legacy `.ppt` files require conversion to `.pptx` before native text extraction. Different recording lengths require per-meeting processing and recoverable chunks. A PPT file alone does not establish slide-display timestamps. Topic relevance and display time must be distinguished, especially when a speaker refers to an earlier slide.

5. Final architecture and implementation

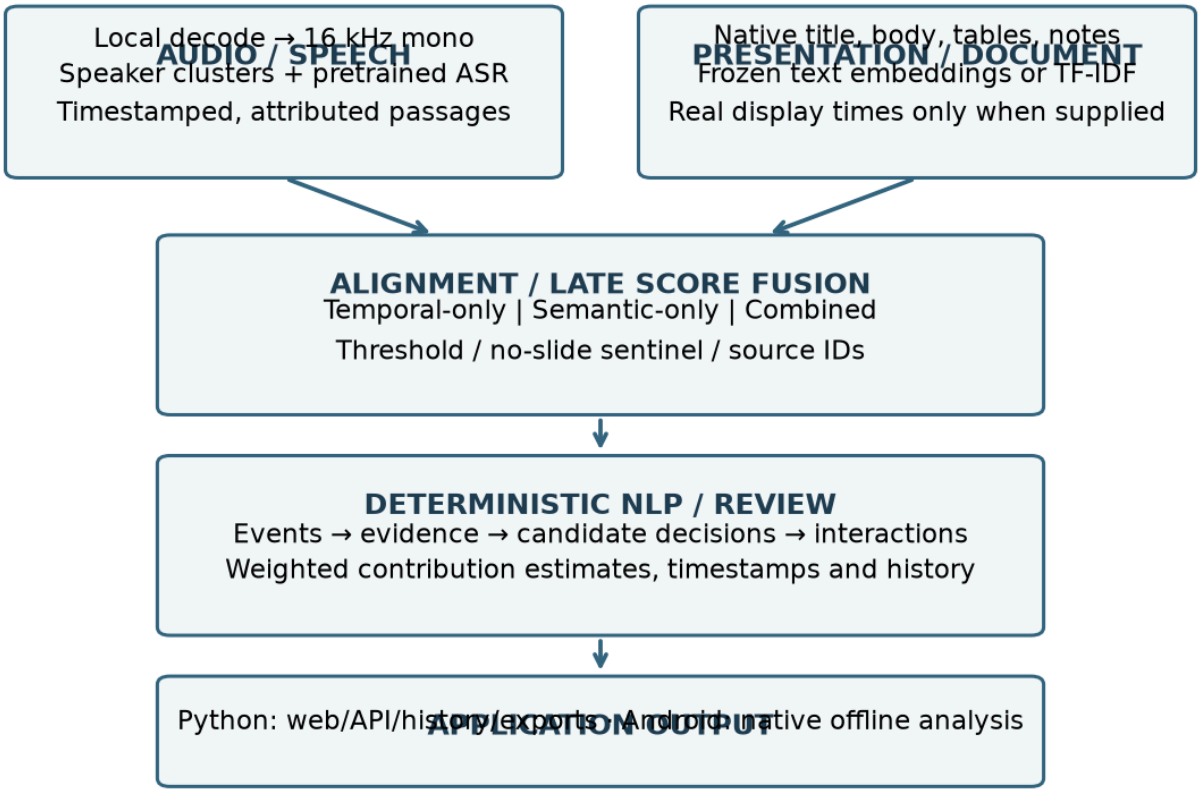


Figure 2. Final processing architecture. The speech branch uses pretrained inference; the document branch extracts native text. Source-linked NLP follows alignment. The Android implementation is lighter than the Python evidence-graph workflow.

5.1 Laptop workflow

Stage	Implemented behavior / source area
Input and decoding	Browser recording/uploads; FFmpeg conversion to normalized WAV; `src/audio/`
Speaker analysis	sherpa-onnx segmentation, embedding and clustering; pyannote remains an optional alternative
Transcription	faster-whisper multilingual small by default, CPU/int8, four threads, 120-second central chunks with overlap
Attribution	Temporal matching between ASR words/passages and speaker intervals; ambiguous assignments can abstain
Document extraction	python-pptx titles/body/tables/notes; `src/ppt/`
Alignment	Temporal overlap plus semantic embeddings or optional TF-IDF; `src/fusion/`
NLP and decisions	Rule-based events, evidence relations, candidate decision reconstruction and interaction analysis

Stage	Implemented behavior / source area
Influence	Configurable seven-feature weighted estimates with supporting event/decision references
Presentation	React interface, FastAPI single-worker queue, SQLite ownership/history and structured exports

Stage artifacts are stored under `data/processed/`; report and evaluation artifacts are under `outputs/`. The queue serializes heavy meeting jobs. Resume state and transcription chunks permit recovery after interruption; changed inputs/settings invalidate relevant caches. Speaker failure in automatic mode produces a warning and `UNKNOWN` transcription rather than invented identities. Required-speaker mode fails when separation is unavailable.

5.2 NLP, lineage and influence

Keyword and bounded contextual rules identify candidate events in English/Hindi/Hinglish text. Evidence relations connect related events and selected slides; candidate decisions group proposal/discussion/evidence/revision/agreement/final-event information. Roles such as proposer, objector and evidence provider describe detected contributions, not verified job titles.

The configured influence estimate is:

```
I(p) = 0.15 P(p) + 0.20 E(p) + 0.15 O(p) + 0.20 R(p)
     + 0.15 D(p) + 0.05 X(p) + 0.10 G(p)
```

`P, E, O, R, D, X, G` are normalized proposal, evidence, objection, revision, decision, interaction and slide-grounded contribution features. Weights are prototype defaults, not trained or human-validated coefficients. Scores can be zero when no qualifying evidence is present. `UNKNOWN` is not treated as an identified person in current participant influence output.

Candidate action items remain source sentences rather than newly generated instructions. Named ownership, deadlines and decision status still require review. A source identifier verifies where an extraction came from; it does not prove the extraction’s meaning or factual correctness.

5.3 Simplified interface and persistence

The final interface emphasizes Record meeting, Upload audio, optional Upload PPT and optional vocabulary hints. Users review transcript, participants, actions, decisions, influence, events and presentation evidence. Speaker names/roles are editable and reflected in transcript/export views. History preserves meetings and edits. Settings hold the profile, automatic Hindi/English language selection and advanced controls.

The website accepts supported recordings and converts them locally; users are not required to pre-convert every input to 16 kHz/mono/16-bit PCM. Automatic language observations and Hindi-safe processing are implemented, but natural code-switching accuracy has not been measured. Generative notes/questions and cross-meeting word search were removed from the final product scope.

6. Fusion terminology, representation and formulas

6.1 What fusion is actually implemented

The original inputs are **meeting audio** and **presentation/document content**. The audio branch produces timestamped speaker-attributed text; the document branch produces extracted slide text and embeddings. Temporal and semantic scores connect these representations. They are two alignment signals, not additional independent input modalities.

Implemented technique	Actual code	How it is used
Temporal alignment	`src/fusion/temporal_alignment.py`: `compute_temporal_scores()`	Scores passage/display-interval overlap when genuine slide times are supplied
Semantic alignment	`src/fusion/semantic_alignment.py`: `compute_semantic_scores()` / `iter_semantic_scores()`	Matches audio-derived text to slide text using a shared frozen encoder and cosine similarity
Combined temporal–semantic alignment	`src/fusion/multimodal_representation.py`: `combine_scores()` / `align_segment()` / `run_alignment()`	Combines available scores, applies candidate filtering/thresholds and returns source-linked records

The application normally uses the combined aligner; for an untimed PPT it correctly falls back to content similarity. The assessment runner isolates temporal-only and semantic-only to compare their contributions. These are three executable alignment configurations of the implemented components, not three separate trained models or three user-facing model-selection options.

The fused record preserves passage ID, speaker, start/end, transcript text, selected slide and time/content/combined scores. Event extraction carries the slide ID onward, evidence links refer to the slide/passage, and candidate decisions/influence retain supporting source IDs. This propagation is the project's implemented multimodal integration beyond transcription.

Optional lexical fusion uses `src/fusion/lexical_alignment.py`: `TfidfIndex`. It replaces the content scorer in the Python path; native Android uses its own slide-text TF-IDF implementation. No visual encoder, OCR or acoustic–image attention is claimed.

6.2 Early, intermediate and late terminology

The fusion literature distinguishes the stage at which information is combined [2]. In this project, implementation status must accompany the taxonomy:

Fusion family	Typical combination point	Status in this project
Early / feature fusion	Combine modality features before a shared predictor	No separately trained early-fusion classifier/network is deployed
Intermediate / joint-representation fusion	Combine encoded representations inside a joint learned model	No joint acoustic–visual encoder or cross-attention model is deployed

Fusion family	Typical combination point	Status in this project
Late / score or decision fusion	Combine independently computed outputs/scores	Implemented: content similarity and optional display-time overlap are combined

A joined JSON record of speaker, text and slide ID is not by itself a trained early-fusion model. The temporal–semantic combination is a hybrid of two alignment signals, but is still **late score-level fusion** in this taxonomy. Calling it “hybrid” does not establish a learned intermediate architecture. No empirical early-versus-intermediate-versus-late neural ranking is claimed.

6.3 Content and time representations

The slide representation concatenates title, body text, table cells and notes. The Python embedding model is all-MiniLM-L6-v2, producing 384-dimensional text vectors [6]. The same frozen encoder processes transcript passages. Its output expresses relevance, not entailment, and has not been validated here for Hindi/English cross-language matching.

For passage `i = [a_i, b_i]` and displayed slide interval `j = [s_j, e_j]`:

```
Overlap(i,j) = max(0, min(b_i,e_j) - max(a_i,s_j))
T(i,j)       = min(1, Overlap(i,j) / max(b_i-a_i, epsilon))
S(i,j)       = dot(u_i,v_j) / (norm(u_i) * norm(v_j))
C(i,j)       = (w_t*T(i,j) + w_s*S(i,j)) / (w_t+w_s)
```

`T` is unavailable when a slide has no valid start/end time. `S` is cosine similarity and can be negative; it is not a probability. Empty text uses a zero vector. Defaults are `w_t = w_s = 0.5`, top-k = 3 and selection threshold 0.3. With missing timing, the deployed combined path falls back to content similarity. With timed overlapping slides, the current candidate filter can exclude offscreen slides before scoring. Low-score results are marked `NO_CONFIDENT_SLIDE`.

6.4 Three configurations compared

Configuration	Information used	Main advantage	Main limitation
Temporal-only	Display overlap, no content score	Resolves context-dependent speech when the active slide is known	No basis for untimed decks; can link unrelated speech to the active slide
Semantic-only	Passage–slide embedding similarity; timing/filter removed	Supports paraphrases, untimed slides and offscreen references	Generic speech can lack a clear semantic match
Temporal–semantic hybrid	Current timed candidate filter and weighted time/content score	Combines contextual display evidence with content matching	Hard timing filter can suppress valid offscreen matches; time can overwhelm abstention

The semantic-only experiment removes the timing filter completely. Merely setting the temporal weight to zero while retaining that filter would still use time and would not be a clean semantic-only baseline. Temporal-only uses zero content scores and abstains for untimed inputs.

A fourth, optional deployment scorer is TF-IDF lexical similarity. It replaces the semantic content score; it is not ensembled with embeddings in the current pipeline. Android uses native-text TF-IDF. TF-IDF is a useful lightweight alternative but its accuracy is not measured by the three embedding/time experiments in this report.

7. Experimental design and metrics

7.1 Controlled diagnostic

The final assessment runs 24 constructed transcript cases against six constructed slides. The slides cover budget, performance, schedule, offline deployment, risks and participant responsibilities. Six groups contain four cases each: exact content, paraphrases, offscreen references, missing timestamps, context-dependent statements and unrelated/blank speech.

Expected relevant slide IDs or a no-slide label are specified in the authored fixture. Cases with times use a 5–10 second passage, a designated displayed slide from 0–30 seconds and other slides from 40–70 seconds. Untimed cases supply no slide times. All methods use the same cases, slide vectors and threshold 0.3, with a frozen encoder and four CPU threads. No weights are fitted to these cases and there is no train/validation/test split. This is an illustrative regression/behavior experiment, **not an independently annotated meeting benchmark**.

The target is relevant slide content, which can differ from the displayed slide during an offscreen reference. Ground-truth author intent is known by construction. This makes the timed filter's limitations visible but does not estimate their frequency in real meetings.

7.2 Metric definitions

```
Top-1 correctness = correctly selected slide/no-slide cases / all cases
Coverage          = selected non-sentinel cases / all cases
Grounded precision = correct slide links / all predicted slide links
Grounded recall    = correct slide links / all cases needing a slide
Grounded F1        = 2*precision*recall / (precision+recall)
WER                = (substitutions+deletions+insertions) / reference words
DER                = (missed speech+false alarm+speaker confusion)
                     / reference speaker-time, after label mapping
RTF                = processing time / audio duration
```

Coverage is not accuracy: linking every passage can create false positives. Exact no-slide matches contribute to top-1 correctness, while grounded precision/recall score actual slide links. CER/event/decision F1/influence correlation require different annotations and are not inferred from these metrics.

7.3 Real unlabelled comparison

The same three alignment configurations are rerun on the saved 2,560.569-second meeting `meeting_8e464f9471b34c46ad9f4ab3a547c923`: 1,008 transcript passages and 34 slides. This is a saved application recording and deck, not an independently labelled slide-relevance benchmark. None of its slides has valid display start/end times. Input hashes and counts are retained, but recording content and personal identities are not reproduced in this report.

The experiment reuses saved ASR/speaker outputs to isolate alignment. It measures coverage, selection agreement and selection-stage runtime. WER, DER and influence are not recalculated for this unlabelled recording. Model loading/encoding are reported separately from selection; timings are one local run and not a general performance ranking.

8. Results: comparison of all three alignment techniques

8.1 Controlled overall results

Method	Correct / 24	Top-1	Grounded precision	Grounded recall	Grounded F1	Coverage
Temporal-only	12	50.00%	60.00%	60.00%	0.6000	83.33%
Semantic-only	20	83.33%	100.00%	80.00%	0.8889	66.67%
Temporal–semantic hybrid	16	66.67%	66.67%	80.00%	0.7273	100.00%

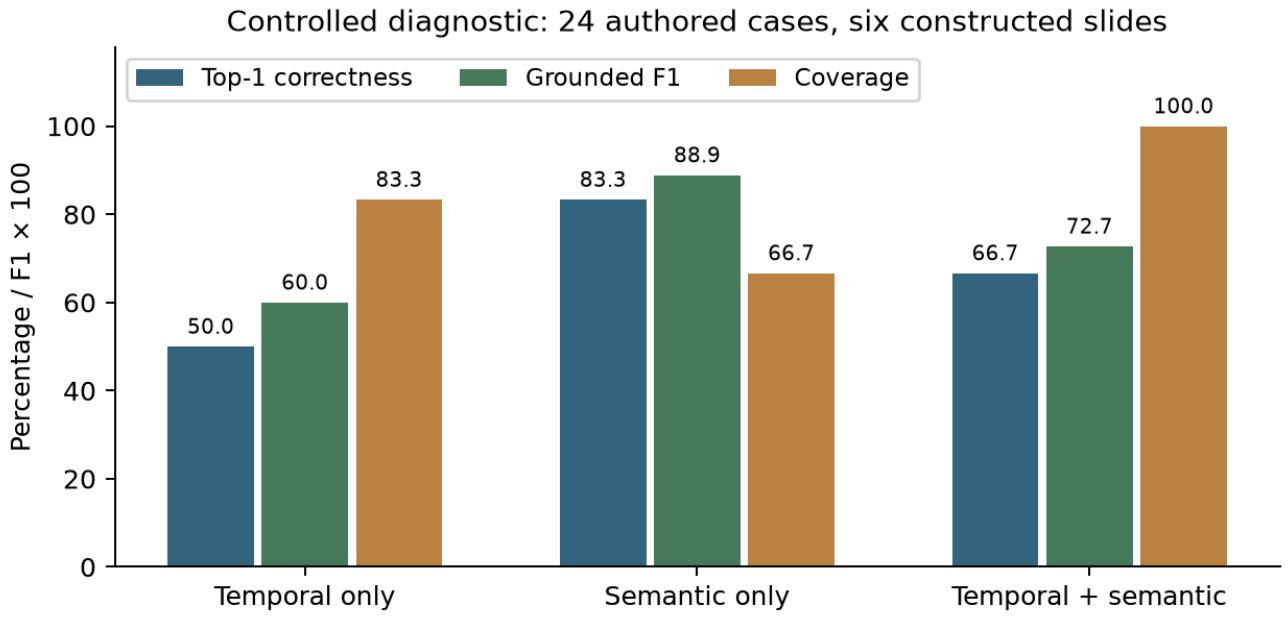


Figure 3. Results on 24 author-designed cases. These are controlled diagnostic scores, not AMI slide-alignment accuracy or evidence of a statistically generalizable winner.

Semantic-only performed best on the chosen diagnostic cases. Its lower coverage reflects appropriate abstention on unrelated/blank speech and failure on generic context-dependent statements. The combined method recovered generic timed statements, but selected incorrect active slides during offscreen references and failed to abstain on unrelated speech. Temporal-only was unable to handle untimed slides or distinguish relevant from unrelated speech within a display interval.

8.2 Results by case type

Case type, four cases each	Temporal-only	Semantic-only	Hybrid
Exact content	4/4	4/4	4/4
Paraphrases	4/4	4/4	4/4
Offscreen references	0/4	4/4	0/4
Missing slide times	0/4	4/4	4/4

Case type, four cases each	Temporal-only	Semantic-only	Hybrid
Context-dependent statements	4/4	0/4	4/4
Unrelated/blank speech	0/4	4/4	0/4

Model loading took 1.4648 seconds and shared slide/passage encoding took 0.2559 seconds for the controlled fixture. Those costs are distinct from selection time; temporal-only can operate without a text encoder.

No statistical confidence interval or significance claim is reported for this small constructed fixture. Its purpose is to demonstrate the three executable strategies and reveal concrete behaviors, including failures. The per-case inputs, intended labels and outputs are included with the report assets for inspection.

8.3 Real unlabelled results

Method	Linked / 1,008 passages	Coverage	Selection-stage time	Alignment accuracy/F1
Temporal-only	0	0.00%	0.0321 s	Unavailable
Semantic-only	39	3.87%	0.0442 s	Unavailable
Temporal–semantic hybrid	39	3.87%	0.0360 s	Unavailable

Shared passage encoding took 4.6531 seconds in this run; the table times cover selection only and exclude that encoding and model loading.

Semantic/hybrid selected the same result for **100% of passages** because all slides lacked display times. That equality validates the fallback; it is not a benefit from temporal fusion. Low 3.87% coverage needs investigation against independently labelled relevance: it can reflect deck/recording mismatch, threshold/model behavior or largely unrelated discussion. Without labels, it cannot be interpreted as 3.87% accuracy or 96.13% failure.

8.4 What the comparison supports

Content information is necessary for untimed slides and offscreen references. Display context can help generic statements but can also generate false links. The current combined scheme is not universally better. Future experiments should test a semantic relevance gate before using time to boost scores, and retain offscreen candidates when appropriate. These are proposed changes, not deployed or benchmarked improvements. Real timed, paired recordings and independently checked slide labels are required before selecting a generally best configuration.

9. Speech-recognition and speaker results

9.1 Laptop English AMI measurements

Existing experiments use AMI ES2002a Array1-01, source window 30–630 seconds, duration 600 seconds, with manual AMI word annotations and normalized full-transcript WER. The window includes some pre-meeting material not fully represented in the reference. Different runs/settings/cache states mean timings are descriptive rather than controlled speed comparisons.

Saved experiment	WER	Elapsed	Peak RSS	Scope / difference
Initial small	45.84%	174.973 s	890.1 MiB	Historical small-model run
Initial medium	66.76%	329.049 s	3,008.0 MiB	Earlier poorer medium result retained
Medium, language scope adjusted	40.83%	357.776 s	2,309.2 MiB	Changed run/settings; not cherry-picked as an identical comparison
Small, 300-second chunks	45.84%	82.254 s	1,516.7 MiB	Throughput profile
Small, 120-second chunks	41.49%	115.043 s	1,479.7 MiB	Different chunk profile
Later small, 120-second chunks	42.30%	105.945 s	1,057.2 MiB	Later saved website-scope benchmark

The 120-second profile was slower than 300 seconds in the recorded pair but had lower WER. The latest small result is slightly worse than the earlier 120-second WER. No monotonic accuracy improvement is claimed. These runs used unavailable/disabled speaker separation, so they do not measure complete diarized-transcript quality. WER is an error rate, not a generic application “accuracy” or an influence score.

9.2 Android English AMI measurements

The Android annotated experiment uses the different ES2002a Array1-01 window 60–150 seconds, 90 seconds total, a multilingual base GGML model and known count four. The normalized audio was verified against the source. Manual AMI transcripts and speaker intervals provide the references.

Measure	Saved result
Reference words	265
Substitutions / deletions / insertions	56 / 37 / 7
WER	37.74%

Measure	Saved result
Native speaker-turn DER	13.48%
Delivered passage-attribution DER	13.98%
DER collar	0.25 s
Cluster-label mapping / overlap	Optimal mapping; overlap included

These are emulator-path results, not physical-phone performance. The native speaker test produced four clusters and 12 turns. The Android application test produced 33 attributed transcript passages, six event candidates and no decision candidates; all four participant influence scores were zero. It used three artificial slides, with eight selected links, solely to verify integration. Synthetic-slide counts do not constitute alignment accuracy.

The laptop and Android results use different windows, models and pipelines. Neither platform is declared more accurate on that basis. Natural Hindi/Hinglish, owner/deadline extraction, decision quality and influence agreement remain unmeasured.

10. Historical CA3 outputs and error analysis

The representative CA3 record `meeting_288ab746` remains available. It demonstrates the earlier analytical output, not a independently scored benchmark:

Historical output field	Value
Meeting duration	2,133.02 s / 00:35:33
Reported participant rows	6: five speaker IDs and one UNKNOWN row
Slides	8
Extracted events	162
Decision candidates marked confirmed	1
Action candidates	3
Participant interactions	17

`UNKNOWN` does not mean a sixth identified person. The current implementation excludes unknown identity from identified participant influence. The candidate “Either whatever we decide the default to be” is labelled confirmed by rules, but the sentence fragment alone is insufficient to independently establish a final meeting decision. Candidate status must be checked with its surrounding source events.

10.1 Participation and influence fields

Speaker	Historical duration field	Duration rank	Influence estimate	Influence rank
SPEAKER_04	141.8 s	1	0.3250	2
SPEAKER_00	70.2 s	2	0.3667	1
SPEAKER_05	56.0 s	3	0.0833	5
SPEAKER_02	50.0 s	4	0.1333	3
UNKNOWN	6.0 s	5	0.0167	6
SPEAKER_01	3.0 s	6	0.1167	4

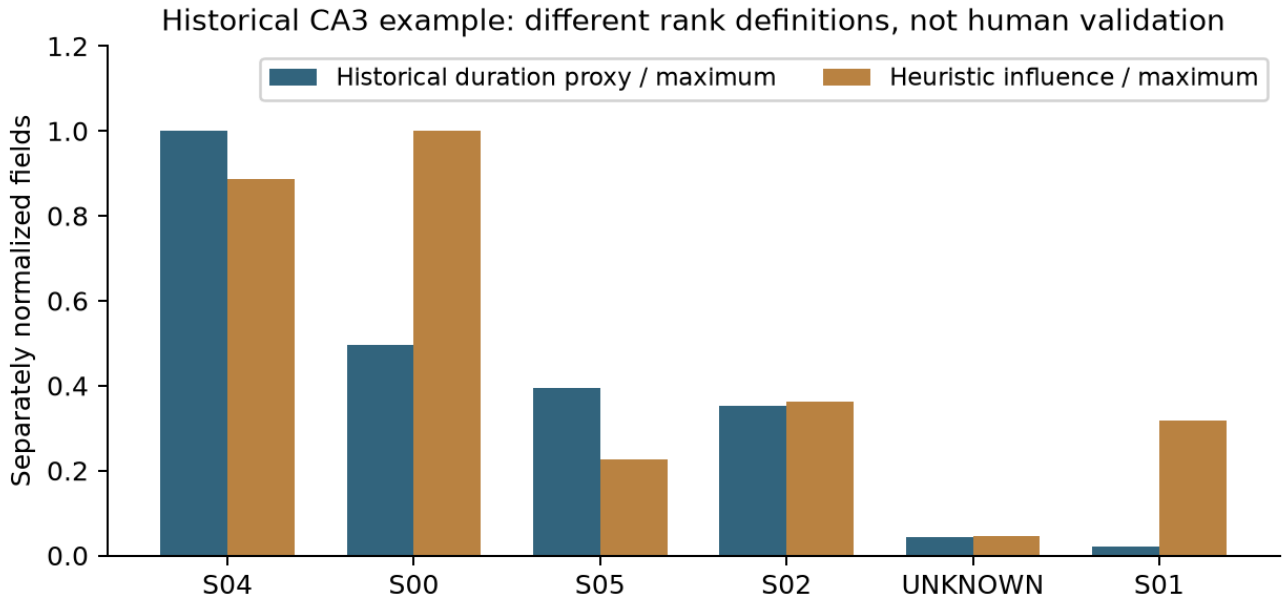


Figure 4. Historical CA3 fields normalized separately for display. The duration field can reflect the older event-duration proxy, not verified total speaking time. UNKNOWN is a historical unresolved row. Different ranks demonstrate different scoring definitions, not proven human influence.

The difference between duration rank and influence rank illustrates how the heuristic uses decision-linked features. It does not establish agreement with human judgement. The earlier statement that one participant's short contribution proved more influence than another's long contribution should therefore be interpreted as a model-output observation.

10.2 Action-item examples

Earlier candidate text	Interpretation in the final assessment
"I'm just going to hopefully add something by next week…"	Plausible task candidate; owner/commitment/deadline still require review
"I'm not afraid to take that."	Not enough evidence by itself to establish a task
"Like take care of yourself."	Conversational phrase; illustrates a keyword false-positive risk

Extracting an exact ASR sentence does not make its category correct. Final wording replaces earlier "100% traceable/non-hallucinating/production-grade" claims with **source-referenced output requiring review**. No custom Whisper pretraining on AMI took place; all speech/speaker/embedding models were used as pretrained inference components.

11. Android, offline execution and optional workstation deployment

The native Android 0.6 prototype is implemented in **Java with C++ inference integration**, rather than the Kotlin wording in CA3. It records 16 kHz mono PCM, accepts supported audio via local platform codecs, imports the speech model, performs speaker analysis, lets users edit speaker names/roles, and extracts candidate NLP/decision/influence results. Native PPT text is indexed with TF-IDF. History, playback, exports and resume checkpoints stay in app-private storage.

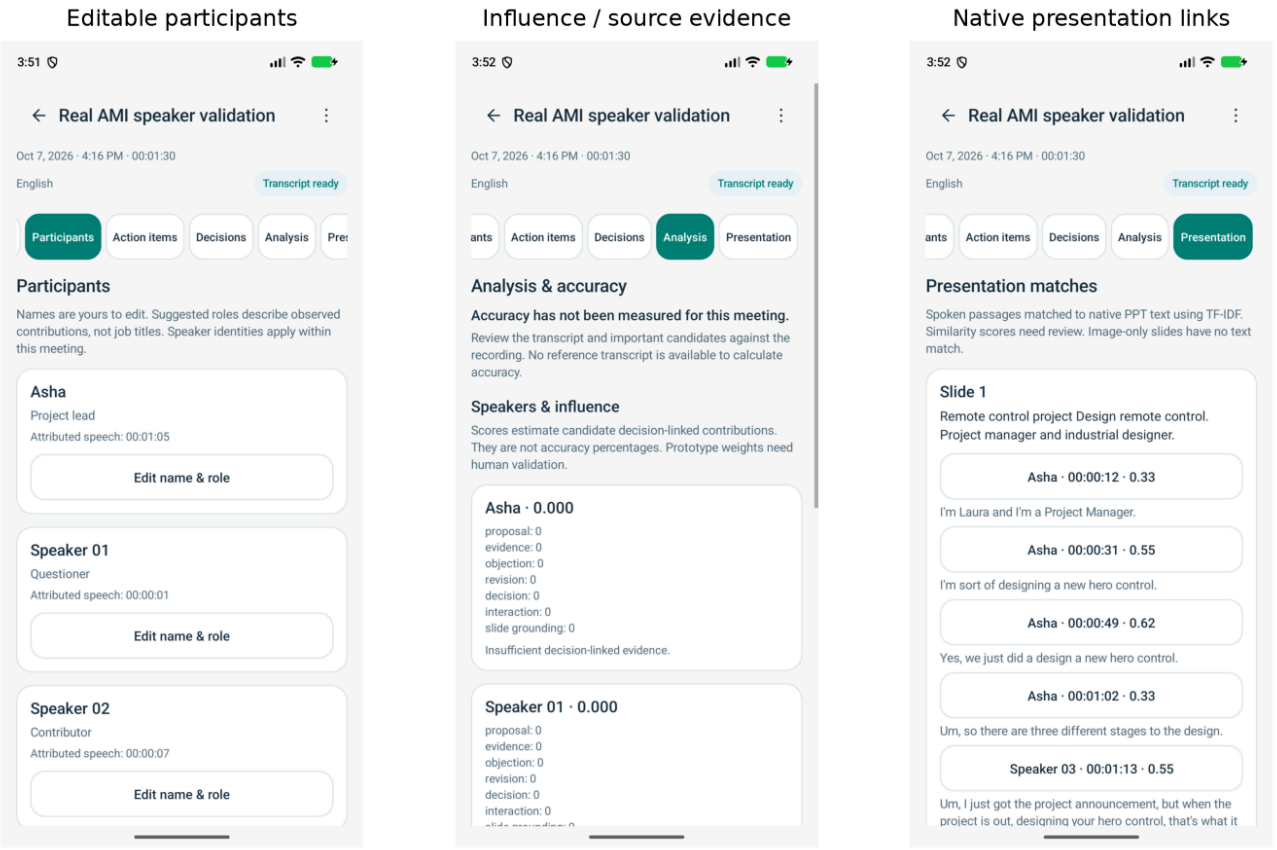


Figure 5. Existing emulator validation screens for transcript/participants, influence and presentation references. Displayed results are examples, not an independent accuracy evaluation.

The app has no INTERNET permission for its native workflow. Model setup and distribution need a provisioned GGML model; speaker weights are packaged. Recording supports up to six hours, but native speaker analysis has a separate one-hour limit. These limits must not be conflated. Pause waits for the current native stage/chunk to finish rather than interrupting inference instantaneously. ARM64 physical-phone battery/thermal/vendor-codec and long-meeting performance are not yet validated.

The Python pipeline can be served on a workstation through the prepared launcher and HTTPS reverse-proxy examples. GPU use requires a compatible CUDA/cuDNN/CTranslate2 environment; an environment variable alone does not install it. Android's optional cloud entry opens a browser site with separate server history; native upload/history synchronization is not implemented. No live public hosting or measured cloud-speed result is reported.

Local processing is useful for confidential institutional/project meetings, poor connectivity, classroom project discussions and controlled research review. Server processing can help users with limited local compute but introduces network, deployment and access-management requirements. Such suitability is an intended use, not evidence of production certification.

12. Processing efficiency and reliability

12.1 Avoided work

The final engineering update shares one cached configured Whisper model in the single worker, shares one sentence-embedding model between slides and passages, batches passage encoding in groups of 32, and normalizes slide vectors once. It also skips rebuilding an unchanged website at startup and records stage timings. Models, beam settings, chunk profile and clustering threshold were not changed by those performance fixes.

A saved warm-model benchmark of 256 real passages and 34 slides measured scalar semantic scoring at **1.6661 seconds** and batched scoring at **1.1672 seconds**, a **1.4274**× stage speedup. All semantic-only selections agreed; rounded scores differed by at most approximately 1.01e-6. This is one local scoring run, not an end-to-end speedup or an alignment-accuracy result.

12.2 Fresh input and cached repeat

A real 90-second AMI integration with four expected speakers and an artificial three-slide PPT produced four participants and 20 Python transcript passages. Its recorded stages were:

Stage	Fresh inference caches	Same-ID cached repeat
Preprocessing	0.0496 s	0.0553 s
Speaker analysis	18.4100 s	0.0441 s
Transcription	16.1251 s	0.0180 s
Slide extraction / embeddings	1.7024 s	0.0327 s
Alignment	0.1282 s	0.1166 s
Remaining NLP / report stages	Approximately 0.043 s	Approximately 0.043 s
Recorded processing elapsed	**36.4631 s**	**0.3138 s**

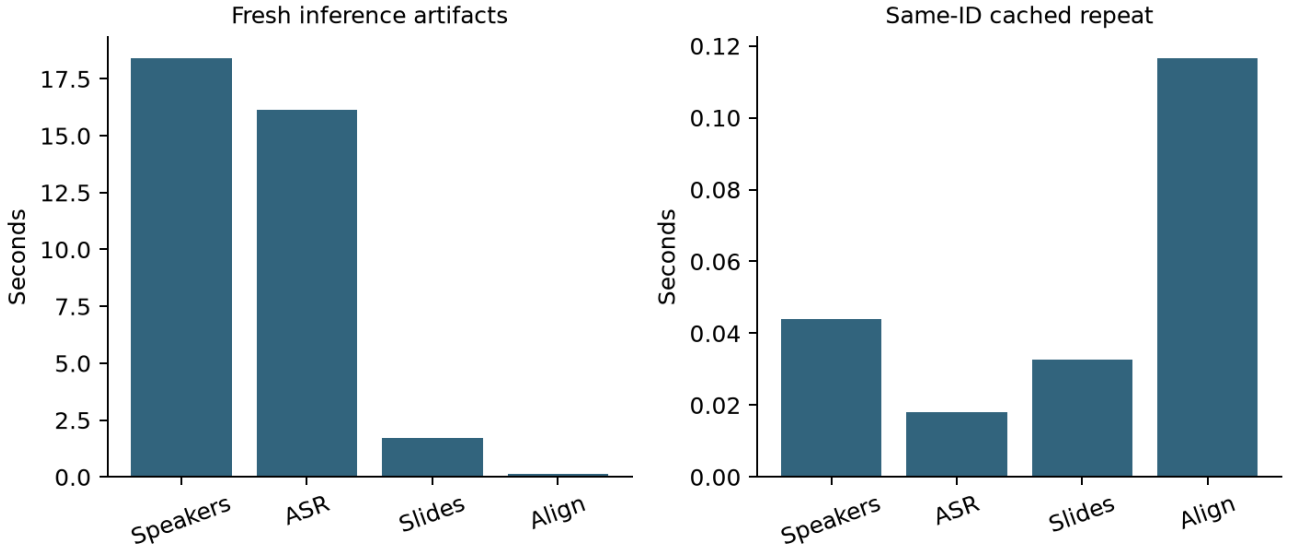


Figure 6. Recorded processing stages for a fresh-ID run and its cached repeat. The second run reuses outputs; it does not transcribe 90 seconds of new audio in 0.31 seconds.

Times exclude module imports, upload/decode, queue wait and final JSON/checkpoint writes. The environment/model files were already provisioned; “fresh” means inference artifacts for that meeting ID were fresh. No matched complete-pipeline before/after benchmark was collected for the optimization.

Separate 90-second speaker threading tests measured 20.4257 seconds at four threads, 35.2162 at one and 26.6317 at two, with the same DER 13.48%. Four remains the default. The cases ran in that order on one file, so this is a local setting check rather than a universal CPU recommendation.

12.3 Historical slowdown diagnosis

Saved comparable-duration 42.7-minute jobs differed in scope: the earlier job lacked successful speaker separation and PPT fusion; the later job performed both. Historical stage timings were absent, so their exact slowdown is not recoverable. New diagnostics distinguish stages, cached chunks and worker progress. Queue wait, fresh model setup, inference and cached reconstruction must be reported separately. Folder cleanup reclaims storage and reduces accidental scanning; it does not accelerate neural inference inherently.

13. Training, baselines and evaluation status

Pretrained perception models do not require retraining to run the app. The project did not train Whisper, diarization weights or MiniLM on AMI. A CPU training CLI exists to fit nonnegative influence-feature weights from human ratings with disjoint meeting-level train/validation/test splits, and exports a research artifact. No real labelled training run or deployment of trained influence weights has been completed.

The evaluation library contains speaking-time/event-count/evidence/transcript-only/multimodal baselines and influence weight-sensitivity experiments. Some inherited baseline implementations use extracted-event durations, while the current main influence runner can use full transcript coverage. Their definitions must be harmonized before publication.

Inherited ablation labels such as “without temporal” or “without semantic” change influence-feature weights rather than rerunning actual upstream fusion. They must not be used as the three fusion comparisons. The CA4 alignment script instead directly reruns the temporal-only, genuinely unfiltered semantic-only and current combined strategies on identical inputs.

The corrected research runner now loads full transcript, diarization and alignment artifacts; resolves decision times from source events; records missing prediction artifacts as errors when references exist; and writes meeting-specific comparison CSVs. Missing human annotations remain explicitly unavailable. Existing matching primitives are coarse for some event/decision cases and do not establish independent human influence agreement.

Evaluation target	Evidence currently available
English ASR	Scoped annotated AMI WER experiments
Speaker attribution	Scoped 90-second AMI DER experiment
Three alignment configurations	Controlled fixture correctness/F1; real unlabelled coverage
Natural timed-slide alignment	Not independently evaluated
Event / decision semantic quality	Candidate outputs and regression checks; no human-scored dataset result
Influence ranking agreement	No independent human ranking study
Natural Hindi/Hinglish	Language/rule implementation; no measured end-to-end accuracy
Physical-phone / cloud performance	No corresponding real-device/deployed benchmark

An adequate next study would annotate genuine paired audio/decks/display times, use at least two raters for event/decision/influence judgement, report agreement, keep meetings/participants or design series disjoint across tuning/test, and evaluate all methods on the same inputs/hardware. Counterfactual causality should not be inferred from heuristic weights.

14. Software validation, resources and risk management

The project runs in an isolated Python 3.11.17 environment with CPU PyTorch 2.5.1, faster-whisper 1.2.1, sherpa-onnx 1.13.8 and sentence-transformers 3.4.1. Direct dependency specifications and a lock snapshot are retained under `environment/`. Model/license notices and explicit provisioning scripts support reproducibility.

Check	Evidence and interpretation
Python regression suite	507 tests passed; seven existing upstream deprecation warnings. Rechecked for final report preparation.
Three-way comparison script	Executed local models and existing aligner; self-checks validate case IDs, blank vectors and missing-time fallback
Website	Existing lint/build and browser checks at widths 360/390/960, with speaker edits/history/evidence/export workflows
API startup	Actual relocated launcher/API health checked with isolated state; server stopped after validation
Android	Existing JVM, build and emulator validation; not a physical-phone certification
Preservation	Older archive hashes, submitted-report content and existing account/history rows verified during reorganization

Software tests establish implementation behavior. They are not statistical research accuracy. Browser fixture responses are intercepted and do not evaluate recognition.

Risk carried forward from Report 2	Implemented mitigation	Remaining boundary
Computation / long files	CPU int8, recoverable ASR chunks, one worker, cached artifacts	Speaker clustering reads whole audio; native speaker limit differs from recording limit
Speech noise / overlap	VAD/diarization and conservative attribution	Incorrect text/speaker labels still propagate
Format compatibility	Local audio decoding and legacy PPT conversion tools	Scanned/visual-only slides are not semantically understood
Cross-modal mismatch	Separate time/content scores, thresholded matches	Hard timing filter and false slide links need evaluation
Misinterpreted analysis	Source references, review warnings and editable speaker metadata	Candidate roles/influence/status remain heuristic
Connectivity	Provisioned local inference and native offline app	Optional server requires connectivity and deployment preparation

Risk carried forward from Report 2	Implemented mitigation	Remaining boundary
Data persistence	Private ownership checks, history, resume state and backups	Release/backup/access policy must be defined for a real deployment
Research validity	Scoped metrics, input hashes, explicit unavailable labels	Independent meeting-level semantic/influence evaluation is missing

15. Project evolution and cumulative deliverables

Phase	Original aim	Final delivered status
CA1	Literature, problem, architecture and evaluation plan	Problem and modular design retained; implementation scope clarified
CA2	AMI, modalities, resources, risks and EDA	Inputs/provisioning established; current dialogue-act counts corrected
CA3	First decisions/actions/influence output	Historical artifacts retained and independently inspected for false-positive risks
Laptop integration	Usable local application	Recording/upload, private history, editable speaker labels/roles, NLP views and exports
Speaker / native extension	Real participant analysis and PPT support	Local speaker backend and Android speakers/influence/native-text fusion
Reliability / structure	Recoverable and understandable project	Organized folders, categorized handbook, corrected evaluator, batching and stage timings
CA4	Results and three-technique comparison	Authored-case experiment and real unlabelled coverage, plus scoped speech/performance evidence

The older supplied source already contained temporal/semantic alignment, event/evidence rules, candidate decision lineage and weighted influence. These modules are not all new contributions of the laptop migration. The older snapshot lacked a complete live GPU-workstation environment, model weights and dataset, so exact parity cannot be independently established.

The initial source audit compared 147 first-party files (49 identical, 32 modified and 66 current-only), with 561 common report files identical. These are dated snapshot counts. Subsequent application/Android/reporting changes are described here. The redundant older working copy was removed only after verifying a 662-file source/report archive and confirming the old Git commit remained in current ancestry. Existing users/history and the original submitted report were retained.

The cumulative deliverables include the web/API prototype, native Android debug APK, local perception weights, source-linked NLP/report modules, comparison/evaluation/training tools, measured output artifacts, categorized developer/research documentation and this final group report. Debug APK availability is not a signed production/store release.

16. Limitations, future work and conclusion

16.1 Limits of the completed project

The prototype processes recordings, attributes speakers, extracts native slide text, compares alignment configurations and produces inspectable candidate meeting analysis. Its limits are concrete: no learned visual encoder/OCR; no early/intermediate trained fusion network; no natural code-switching benchmark; no independently scored genuine timed-slide dataset; no human-validated influence model; and no physical-phone/cloud deployment benchmark. Global clustering still loads an entire waveform. Keyword event/decision/action extraction can misclassify conversational fragments.

The controlled fixture is deliberately small and illustrative. Its best-performing method should not be promoted as a general winner. The real untimed deck cannot demonstrate temporal fusion accuracy. Single AMI windows do not establish cross-language or cross-meeting generalization. Historical event-duration participation fields cannot be treated as ground-truth airtime, and inferred decision status cannot be treated as a verified meeting outcome.

16.2 Prioritized next work

1. Collect genuine paired recordings/decks/display times and independent relevant-slide/no-slide labels.
2. Test a relevance gate and offscreen-candidate retention, then rerun the same held-out cases without tuning on test labels.
3. Add human-labelled natural Hindi/English/Hinglish meetings and report language-specific WER/CER/DER and semantic extraction metrics.
4. Annotate decision contributions with independent raters, use tie-aware ranking evaluation and fit weights only on disjoint training meetings.
5. Validate ARM64 phones for thermal/battery/codec/interruption and long-recording behavior.
6. Evaluate a provisioned workstation with network/queue/cold/warm latency separated, if remote deployment is selected.

16.3 Conclusion

The project progressed from an architecture and literature proposal to a working local meeting-analysis system with two application paths and reproducible evidence. Audio-derived transcripts and presentation text are linked by temporal/content scoring, then carried into structured event, decision and participant-contribution output. The final three-way comparison shows that each alignment signal has a distinct role: semantic matching handles content and untimed/offscreen slides, temporal matching resolves displayed context, and their present combination trades greater coverage for false-positive and filtering risks.

The main completed contribution is a source-linked offline integration and usable review workflow. The strongest current research finding is the measured behavior of the three alignment configurations under explicit inputs and limitations. General alignment

superiority and valid human influence inference remain hypotheses requiring independently labelled real meetings. The final report records that boundary while preserving the full CA1–CA4 project history.

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[11] Group 12, **Multimodal AI Report CA3**, original submitted PDF, 10 pages; methodology and first application outputs.

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Appendix A. Controlled case manifest

The fixture is fully inspectable in `final_report_assets/evidence/controlled_cases.json`. It contains six slide texts, 24 passage texts, intended relevant slide/no-slide labels, designated displayed slide IDs and a timing-availability flag. The texts are constructed examples, not extracts from private user recordings.

Case IDs	Case family	Intended label policy
01–04	Exact content	Same slide as passage content and displayed context
05–08	Paraphrase	Meaning matches a relevant slide; display agrees
09–12	Offscreen reference	Content refers to an earlier/different slide than the displayed one
13–16	Missing display times	Content label exists; no temporal evidence supplied
17–20	Context-dependent	Author intent targets the displayed slide; generic text is insufficient alone
21–24	Unrelated or blank	Correct output is `NO_CONFIDENT_SLIDE`

Method outputs, counts, model/threshold settings and fixture hash are in `controlled_comparison.json` and the per-method alignment JSONs. A row in this manifest is not an independently human-annotated natural meeting. Dataset construction is part of the reproducible diagnostic definition.

Appendix B. Result provenance and reproducibility

Evidence artifact, project-relative	What it establishes
`outputs/evaluation/final_ca4/controlled_cases.json`	Constructed inputs and authored relevant-slide labels
`outputs/evaluation/final_ca4/controlled_comparison.json`	Three-method controlled metrics and per-group outputs
`outputs/evaluation/final_ca4/real_unlabelled_comparison.json`	Saved real-input hashes, coverage, timing availability and fallback agreement
`outputs/evaluation/final_ca4/eda_inventory.json`	Direct dialogue-act XML counts and local file inventory
`outputs/evaluation/benchmark_*_benchmark.json`	Historical annotated laptop ASR measurements and scope
`outputs/evaluation/android06_ami90_annotated.json`	Annotated Android-path WER/DER, source/report hashes and references
`outputs/evaluation/maintenance_alignment_batch.json`	Scalar/batched semantic-stage timing and score agreement
`outputs/evaluation/maintenance_pipeline_timing.json`	Fresh/cache processing stage durations
`outputs/reports/meeting_288ab746_report.json`	Historical CA3 output fields, not human-verified labels
`logs/final_report_20261010/python_tests.log`	Current regression run
`docs/README.md`	Master setup/developer/research index

The report assets carry compact copies of relevant JSON evidence, figures and a manifest. Full speech-reference JSONs and recordings remain in the local project; their aggregate numbers are reproduced here. Original submitted PDF content is preserved. The optional complete-record PDF appends those historical submissions and the individual review without editing them; earlier defaults/assertions in those appendices are superseded by the main report's explicit findings.

```
cd '/home/amitkumar/Music/MAI Project'
OMP_NUM_THREADS=4 .venv/bin/python scripts/compare_fusion_ca4.py
OMP_NUM_THREADS=4 bash scripts/check.sh -q
.venv/bin/python scripts/build_final_report.py

# To run the application:
bash scripts/start_app.sh
```

Comparison reruns use installed local models and saved inputs; they do not train models or require a hosted LLM. Numerical selection timings vary with machine load. Re-running the report builder reads the current Markdown and evidence files and recreates figures/PDFs; it does not silently regenerate research results. Re-running an experiment first can change results and must be accompanied by an updated report narrative.

Appendix C. Developer handover

```
backend/       FastAPI, local accounts, ownership, uploads and job queue
src/audio/     decode/preprocess, ASR, speaker analysis and attribution
src/ppt/       native presentation extraction and text embeddings
src/fusion/    lexical/semantic/temporal alignment and source-linked records
src/meeting/   events, evidence, decisions, interactions and influence
src/report/    report assembly and analytical/transcript exports
src/evaluation/  metrics, baseline definitions and weight sensitivities
frontend/      React/Vite source and generated website
android/       native Java/C++ app, models/notices, tests and debug APK
environment/   Python requirements and lock snapshot
config/        YAML settings, env template/private env and pytest config
scripts/       setup, startup, checks, provisioning, comparison and PDF build
data/          recordings, annotations, stage artifacts and SQLite state
outputs/       reports, evaluations and research/training artifacts
docs/          categorized guides, research, validation and submissions
archive/       verified older-source/history snapshot
.runtime/      private job state, signing secret and validation tooling
```

Use the project Python 3.11 venv and locked environment. Provision speech, slide and speaker weights explicitly once, then run locally. Configuration is `config/config.yaml`; history is `data/state/nexus_v2.db`; private runtime secrets and uploads should be retained for continuity. Do not publish recordings, account state, credentials or archived desktop configuration with a paper. Advanced YAML changes require a restart. The final report is an evidence-based assessment of a research prototype, not a deployment approval or independently validated journal claim.

MULTIMODAL AI

PROJECT REPORT

"Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis"

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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AY 2026-27

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

CERTIFICATE

This is to certify that the Project work entitled "**Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis**" is carried out by **Anushka Chatterjee, Amitkumar Racha, Sahil Awatramani and Shriyans Sahoo** under the Multimodal AI course under **Bachelor of Technology** in **Artificial Intelligence & Machine Learning**, Symbiosis International (Deemed University), Pune during the academic year 2026-2027.

Name and Signature of the Guide

Dr. Sumit Kumar

Head, Department of AI&ML

DECLARATION

I hereby declare that the project titled "**Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis**" submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Artificial Intelligence & Machine Learning is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

Name(s) of Student(s) : Anushka Chatterjee, Amitkumar Racha, Sahil

Awatramani and Shriyans Sahoo

PRN: 23070126015, 24070126501, 23070126112, 23070126125

Degree: Bachelor of Technology in AI&ML

Department: Artificial Intelligence & Machine Learning

Title of the project: Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis

_____

(Signatures of the Students)

Date: 30 /08 /2026

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ABSTRACT

Meetings are an excellent source of information not only in oral communications but also in the form of the content presented during the meetings for making decisions. This project aims to develop an offline multimodal meeting intelligence framework that uses both the long duration meeting audio and corresponding PPT presentation to analyze and understand the meeting. This meeting intelligence framework will include the step of speaker diarization and automatic speech recognition to generate a transcribed conversation with the attributions of speaker, processing the PPT content, alignment of the audio content with the corresponding slides, extracting the proposals, objections, evidence, revisions, decision and action items to capture decision lineage. Lastly, the participant contribution analysis is done in order to capture decision-related influence of participants considering their evidence, interaction and impact on decisions apart from speaking time.

Keywords: Multimodal Meeting Intelligence, Offline AI, Speaker Diarization, Automatic Speech Recognition (ASR), PPT Analysis, Audio-Visual Alignment, Decision Lineage, Decision Mining, Action Item Extraction, Participant Contribution Analysis, Evidence Extraction, Meeting Analytics, Multimodal Fusion, Natural Language Processing (NLP)

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Chapter 1

Introduction

1.1 Introduction

Meetings continue to be one of the key channels of communication through which organisations disseminate information, deliberate possibilities and make decisions, while the artefact created as a result, such as the audio recording and, usually, the slide presentation, tends not to be subjected to any analysis after the meeting is completed. Most currently available software only allows one to generate a transcript or some sort of abstracted overview without making the link between the spoken content and the visuals and understanding why some specific decisions have been made by the participants of the meeting explicit in the process. The goal of this project is to design a privacy-protecting and offline Multimodal Meeting Intelligence System, which will allow to process simultaneously audio recordings of the meeting and the slides and to generate a speaker-attributed transcription of the meeting, to find the correspondence between the spoken content and the evidence presented in the form of slides, to identify inconsistencies between the two modalities and to detect the meeting events such as suggestions, objections, decisions and action items. In addition to this, the system will be able to quantify the level of influence each participant had on the final outcome of the discussion in an explainable way. In order to formulate the problem statement, the goals and the project roadmap, the group undertook a review of the literature on the topic including twenty-eight peer-reviewed sources covering the domains of speech enhancement, streaming and multi-talker recognition, speaker diarization, multimodal meetings datasets and participant role and influence modelling.

1.2 Problem Statement

Many times, an offline meeting has much valuable information available in multiple modalities, including spoken discourse and PPT presentations. Current meeting analysis techniques mainly concentrate on transcription and generic summarization; however, they do not support alignment of

spoken statements to the corresponding presentation content, the decision lineage, and determining participants' contributions to the decisions.

Manual analysis of a one or two-hour meeting together with a PPT presentation is labor-intensive and does not allow finding systematically valuable discussion, proposals, oppositions, evidences, decisions, and action items. In addition, speaking time does not give any insights about the actual participant contribution or influence, as one short statement can change everything.

Thus, the project suggests an offline multimodal meeting intelligence solution that takes into account both meeting audio and the corresponding PPT presentation and allows performing speaker diarization and speech recognition, extracting relevant information from the presentation slides, aligning discussion segments to the relevant slides, detecting important meeting events and decisions, finding inconsistencies between the spoken claims and presented evidences, building decision lineage, and estimating contribution and influence based on evidences.

It is expected that the proposed solution will be able to process long-duration meetings efficiently using local computation resources while preserving the explainable connection of the result analysis to the speaker statements and timestamps and presentation content.

1.3 Motivation

Meeting is an important aspect of organizational decision-making, however, the information generated within the course of the meeting is disseminated into multiple types of communication. The participants use oral communication as they discuss various issues, while the presentation slides contain supplementary information in terms of figures, tables, propositions, and numbers. Traditional meeting transcription and summarization techniques are mostly concerned with the transformation of speech into text and summarizing the discussion, however, they fail to relate the discussion to the slides used in the course of the meeting.

This problem becomes especially relevant in case of offline meetings of considerable duration since

reviewing two or three hours of discussion, as well as the slides, takes a lot of time. Critical propositions, objections, evidence, decisions, and action items could be dispersed throughout the course of the meeting, and, therefore, might be lost in the conventional summary.

Multimodal analysis could give a more comprehensive view by analyzing what was said, by whom, and what slide was being discussed at the moment. The alignment of speech with PPT slides can give additional proof to the extracted claims and point out possible discrepancies between the verbal information and the information provided in the slides.

Moreover, another important reason is that speaking time does not characterize participant influence on the discussion. The participant who spoke a little may have introduced critical evidence, objected to something important, or proposed something to change that will affect the final decision. Hence, the aim of the project is to analyze participants' contribution in the course of the decision-making process and not just the amount of speech.

The system is motivated by the need of the creation of the evidence-grounded and explainable meeting-analysis framework.

1.4 Objectives

The main goals of the planned research include:

1. Construction of the speaker-attributed transcription pipeline that will work with long-term meeting audio recordings and determine speakers.

2. Capturing and representing textual and visual information from PPT slides to enable incorporation of the information about presentation into meeting analysis.

3. Determining the cross-modal alignment method to link discussion fragments with related information from the presentation.

4. Identification of important events in meetings, including suggestions, objections, evidence, corrections, decisions, and tasks.

5. Detection of possible inconsistencies between claims stated during the discussion and the PPT information and justification of this observation with evidence.

6. Meeting participant influence analysis in the decision-making process through participants' contributions, interactions, evidence and decision-making contributions, accompanied by the analysis explanation.

Chapter 2

Literature Review

2.1 Background

Automated analysis of meetings has advanced in many directions, such as speech enhancement, multi-speaker speech recognition, speaker diarization, multimodal meeting processing, dominance and influence analysis of participants, speech-to-slide alignment, and automatic summarization of meetings. Some of the existing works have devised successful methods to find the speakers, transcribe the conversation, process overlapping speech, study the role of the participants, and align the speech content with the slides used by the presenter in the meeting. Some of the datasets such as AMI corpus have also been useful in the study of multimodal meetings. However, these abilities have been studied separately. Thus, the existing literature motivates an integrated approach using the audio data of the meeting and PPT files.

Ref.	Paper	Year	Research Area	Key Contribution	Relevance to Proposed Work	Limitation / Research Gap
[1]	Dementyev et al., *Towards sub-millisecond latency real-time speech enhancement models on hearables*	2025	Speech Enhancement	Investigates highly efficient speech-enhancement models for low-latency processing.	Provides background for improving quality of meeting audio before downstream processing.	Focuses on real-time hearable devices rather than offline meeting analysis.
[2]	Schröter et al., *Deep multi-frame filtering for hearing aids*	2023	Speech Enhancement	Applies deep multi-frame filtering for speech enhancement.	Supports the need for robust speech preprocessing in noisy recordings.	Primarily designed for hearing-aid applications, not meeting intelligence.
[5]	Dahiya & Kumar, *Noise-robust speech recognition with latency-*	2026	Robust ASR	Studies robustness of speech recognition under	Relevant to reliable transcription of long-duration	Does not address speaker influence, PPT alignment, or

	aware performance trade-offs			noise and processing constraints.	meeting recordings.	decision analysis.
[9]	Li et al., *SkiM: Skipping memory LSTM for low-latency real-time continuous speech separation*	2022	Speech Separation	Proposes an efficient approach for continuous speech separation.	Provides background for handling overlapping speakers in meeting audio.	Focuses on low-latency speech separation rather than higher-level meeting understanding.
[11]	Tripathi et al., *End-to-end multi-talker overlapping speech recognition*	2020	Multi-speaker ASR	Investigates recognition of speech from multiple overlapping speakers.	Relevant to meetings where participants may interrupt or speak simultaneously.	Does not perform meeting-level decision or influence analysis.
[12]	Patino et al., *Enhanced low-latency speaker spotting using selective cluster enrichment*	2018	Speaker Identification	Improves speaker spotting through cluster-based techniques.	Supports identification of individual meeting participants.	Focuses on speaker spotting rather than complete meeting understanding.
[13]	Bredin et al., *pyannote.audio: Neural building blocks for speaker diarization*	2020	Speaker Diarization	Provides neural components and pipelines for speaker diarization.	Directly supports the proposed speaker-attributed transcription pipeline.	Primarily addresses diarization rather than participant influence or decision analysis.
[14]	Zhen et al., *Winner takes it all: An efficient overlap-aware hybrid online diarization with partial backtracking mechanism*	2024	Overlap-aware Diarization	Addresses diarization in the presence of overlapping speakers.	Useful for robust speaker attribution in multi-person meetings.	Designed for online diarization and does not address multimodal meeting reasoning.
[15]	Cornell et al., *Overlapped speech detection and speaker counting using distant microphone*	2022	Overlap Detection	Detects overlapping speech and estimates the number of speakers using microphone	Relevant to identifying difficult multi-speaker segments in meeting recordings.	Focuses on speech-level analysis rather than meeting decisions.

	arrays			arrays.		
[16]	Wang et al., *The MISP 2022 challenge: Audio-visual diarization and recognition*	2023	Multimodal Speech Processing	Provides benchmarks for audio-visual diarization and speech recognition.	Demonstrates the value of multimodal information for multi-speaker meeting analysis.	Primarily uses audio-visual modalities; our system focuses on audio + PPT.
[17]	Eliav & Gannot, *Audio-visual approach for multimodal concurrent speaker detection*	2026	Concurrent Speaker Detection	Investigates multimodal detection of simultaneously active speakers.	Supports robust identification of concurrent speakers.	Uses audio-visual information and does not address presentation-grounded analysis.
[19]	Noulas et al., *Multimodal speaker diarization*	2012	Multimodal Diarization	Proposes multimodal methods for identifying and separating participant speech.	Provides foundational support for multimodal speaker attribution.	Does not connect speaker contributions to presentation content or decisions.
[20]	Chung et al., *Who said that? Audio-visual speaker diarisation of real-world meetings*	2019	Meeting Diarization	Investigates speaker diarization in real-world meeting environments.	Highly relevant to reliable identification of participants in meetings.	Focuses on speaker attribution rather than decision-linked influence.
[21]	Ahmad et al., *Multimodal speaker diarization using a pre-trained audio-visual synchronization model*	2019	Multimodal Diarization	Uses pretrained audio-visual synchronization for speaker diarization.	Provides evidence for pretrained multimodal approaches to speaker attribution.	Relies on visual information and does not analyze meeting decisions.
[22]	Carletta et al., *The AMI Meeting Corpus: A pre-announcement*	2006	Meeting Corpus	Introduces a multimodal meeting corpus containing synchronized meeting information and presentation-	Provides an important foundation for research involving meeting audio and presentation content.	Dataset itself does not provide automated decision/influence analysis.

				related data.		
[23]	Hung et al., *Estimating dominance in multi-party meetings using speaker diarization*	2011	Participant Dominance	Estimates participant dominance using speaker-level meeting information.	Directly motivates participant-level analysis beyond simple transcription.	Conversational dominance does not necessarily represent influence on a specific decision.
[24]	Dong et al., *Using the influence model to recognize functional roles in meetings*	2007	Influence & Roles	Applies an influence model to recognize functional participant roles.	Provides a foundation for modelling influence and participant interactions.	Does not explicitly connect influence to evidence and final decisions.
[25]	Garg et al., *Role recognition for meeting participants: An approach based on lexical information and social network analysis*	2008	Roles & Interaction	Combines lexical information and social-network analysis for participant-role recognition.	Supports the use of linguistic content and interaction patterns for participant analysis.	Focuses on role recognition rather than decision-specific influence.
[26]	Anderer et al., *MaViLS, a benchmark dataset for video-to-slide alignment*	2024	Speech–Slide Alignment	Investigates alignment between meeting content and presentation slides using speech, OCR and visual features.	Directly supports the proposed audio–PPT alignment component.	Focuses on video-to-slide alignment rather than decision and influence analysis.
[27]	Krishna et al., *Team Hitachi @ AutoMin 2021: Reference-free automatic minuting pipeline with argument structure construction over topic-based summarization*	2021	Meeting Summarization & Arguments	Develops an automatic minuting pipeline using topic-based summarization and argument structure.	Supports extraction of structured meeting events and argumentative relationships.	Does not integrate PPT evidence with participant influence and decision lineage.

Table 2.1: Summary of Literature Review

2.2 Research Gap

Several methods have been successfully introduced for enhancing speech, speaker diarization, summarization of meetings, analysis of domination and influence of participants, and alignment of speeches with PPT slides. Yet, the above functions are generally researched independently.

One thing that is missing here is a unified system to analyze long-term audio from meetings combined with PPT slides and link speakers' involvement to the discussions and evidence used and decisions made at the meeting.

In particular, the problem with current domination models is that domination in a conversation does not necessarily imply the influence on decisions. The participants might speak less but provide the crucial evidence or suggestions which lead to the altered decision.

Thus, the main goal of this research is the combination of:

Speaker Attribution → Audio-PPT Alignment → Event/Evidence Extraction → Decision Reconstruction → Participant Contribution → Decision Influence

Chapter 3

Methodology

The suggested system will employ an offline multimodal approach having a modular architecture to analyze the long meetings by means of the meeting audio and the associated PPT presentation. The methodology will consist of several processing stages wherein the output from one stage will serve as an input for the next stage.

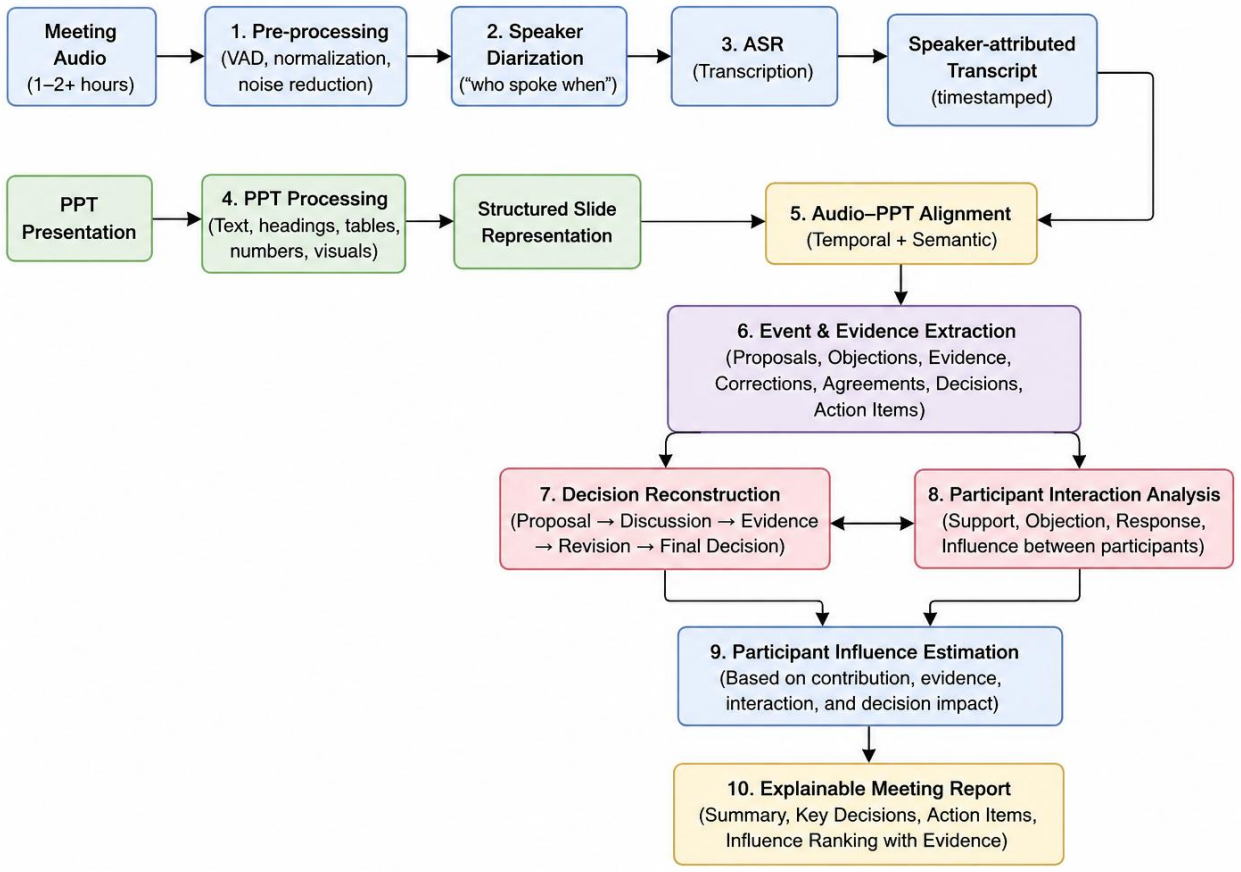


Figure 3.1: Proposed System Architecture

3.1 System Architecture

Layer	Proposed Approach	Expected Output
1. Audio Input & Pre-processing	Accept long-duration meeting recordings and perform resampling, normalization and Voice Activity Detection. Lightweight noise reduction will be applied only when required.	Clean, speech-focused audio segments
2.Speaker Identification	Apply speaker diarization and speaker embeddings to separate participants and maintain consistent speaker identities throughout the meeting.	Timestamped speaker segments
3. Speech Recognition	Transcribe the complete meeting using an offline-capable ASR model while retaining timestamps and speaker IDs.	Speaker-attributed transcript
4. PPT Understanding	Extract text, headings, tables, numerical values and relevant visual information from each PPT slide and generate slide-level representations.	Structured PPT representation
5. Audio–PPT Fusion	Match spoken segments with relevant slides using temporal information and semantic similarity between transcript and slide representations.	Speaker–speech–slide associations
6.Meeting Event Analysis	Identify proposals, questions, objections, supporting evidence, corrections, agreements, decisions and action items from the aligned multimodal information.	Structured meeting events
7. Evidence & Decision Analysis	Associate claims and evidence with relevant slides and reconstruct how proposals evolve through discussion toward final decisions.	Evidence-linked decision chains
8. Interaction Analysis	Model responses, support, objections and influence between participants to determine how individual contributions affect the discussion and decisions.	Participant interaction graph
9. Influence Estimation & Report Generation	Calculate an explainable participant-influence score using decision-relevant contributions rather than speaking time alone, and generate the final meeting report.	Influence ranking + explainable meeting report

Table 3.1: System Architecture Steps

3.2 Processing and Integration

Below are details of the processing stages of the proposed system.

A. Long Audio Processing:

The whole recording will be processed in small parts to make it efficient to process recordings that can take up to an hour or two without keeping all the audio in memory.

B. Speaker Aware Transcription:

ASR output and speaker diarization results will be combined to form a transcript that looks like:

Timestamp → Speaker → Statement

This format will be used as the main input for further analysis.

C. PPT Representation:

Each PPT slide will be transformed to a structured representation containing all the information available from it, both visual and textual. This will allow determining which part of the presentation corresponds to a certain discussion.

D. Multimodal Fusion:

A transcript and PPT slides representations will be merged using temporal and semantic alignment to form an integrated representation containing speaker, statement, timestamp and corresponding slide.

E. Decisions and Influence Analysis:

The fused representation will be analyzed to find events related to the decision-making process as well as the relations between them. Then the participant influence will be estimated depending on the effect of his contribution on the discussion and final decisions.

3.3 Influence Scoring

In contrast to measuring speaking duration, the suggested influence measurement scheme is going to take into account the following contribution signals:

- Participation – speaking turns and relevant participation.
- Contribution to proposal – proposal ideas or proposals made.
- Contribution to evidence – facts or information introduced.
- Contribution to objection/challenge – issues identified or challenges.
- Contribution to decision – the contribution that leads to a change and impacts the decision.
- Contribution to interaction – reactions or support/opposition caused by the contribution of a participant.

The influence scores will be calculated based on the above mentioned combination of signals for each participant and will include the evidence used for scoring to make it explainable.

3.4 Final System Output

The final output will be a structured meeting report containing:

1. Meeting summary
2. Speaker-wise participation
3. Key discussion points
4. PPT-linked statements
5. Proposals, objections and evidence
6. Identified decisions and decision lineage
7. Action items
8. Participant influence ranking with supporting evidence

Chapter 4

Evaluation Plan

The suggested framework will be assessed independently at each important phase as well as at the general analysis of meetings phase. It will help us understand which errors arise due to speech analysis, speaker identification, multimodal alignment, decision extraction, or influence evaluation.

Component	Evaluation Metric	Purpose
Speech Recognition	Word Error Rate (WER)	Measures transcription accuracy
Speaker Diarization	Diarization Error Rate (DER)	Measures accuracy of speaker segmentation and attribution
PPT Alignment	Precision, Recall, F1-score / Alignment Accuracy	Measures whether speech segments are matched with the correct slides
Event Extraction	Precision, Recall, F1-score	Measures identification of proposals, objections, evidence and decisions
Decision Extraction	Precision, Recall, F1-score	Measures accuracy of identifying final decisions
Decision Lineage	Chain/relationship accuracy	Measures whether the system correctly connects proposals, evidence, discussion and decisions
Influence Analysis	Correlation with human ratings / Rank correlation	Measures whether the generated influence ranking agrees with human judgement
Processing Efficiency	Processing time, Real-Time Factor (RTF), memory usage	Measures feasibility for long-duration offline meetings

Table 4.1: Steps of Evaluation Plan

4.1 Baselines

To demonstrate that the proposed multimodal approach provides additional value, its performance will be compared against progressively stronger baselines.

Baseline	Description	Purpose
B1: Transcript-only	Uses ASR transcript without PPT information	Establishes performance using audio/text alone
B2: Speaker + Transcript	Adds speaker diarization to the transcript	Measures the benefit of speaker attribution
B3: Audio + PPT without Decision Analysis	Combines meeting audio and PPT but does not model decision lineage	Measures the contribution of decision modelling
B4: Audio + PPT + Decision Analysis	Includes multimodal alignment and decision reconstruction	Measures the effectiveness of the proposed multimodal decision pipeline
Proposed System	Adds evidence-grounded participant influence analysis	Evaluates the complete system

Table 4.2: Table of Baselines

4.2 Ablation Study

Ablation tests will be performed in order to see what contributes to the overall performance of the system. Various major components such as PPT data, cross-modal alignment, evidence reasoning, interaction modeling, and influence on decision will be removed one at a time, and the performance will be evaluated against the entire system. It will be seen whether there is any improvement with the use of the proposed multimodal components.

4.3 Human Evaluation

In view of the fact that influence cannot be verified from reality, human evaluation becomes more critical. Human raters may be able to independently evaluate selected video clips of the meeting and rate participants based on their influence in decision making. This rating can be compared with that generated by the system. We obtain a much better evaluation than merely stating that the output is a reasonable influence measure.

Chapter 5

Project Timeline

The project will be developed in a phased manner, with each phase building upon the previous stage. The proposed timeline is as follows:

Week(s)	Phase	Key Activities	Deliverable
1–2	**Literature review & problem framing**	Finalize literature review, research gap, problem statement, objectives and evaluation criteria.	Finalized problem definition and literature review
3	**Requirement analysis & architecture design**	Finalize the revised multimodal architecture and select suitable offline/open-source models and tools.	System architecture and tool-selection plan
4–5	**Audio Processing & Speaker Attribution**	Implement audio preprocessing, speaker diarization and speaker-attributed ASR on sample meeting recordings.	Working diarization and transcription pipeline
6	**PPT Processing**	Extract slide text, headings, tables, numerical information and relevant visual features.	Structured PPT representation
7	**Audio–PPT Alignment**	Develop temporal and semantic alignment between transcript segments and relevant PPT slides.	Aligned audio–PPT representation
8	**Event & Evidence Extraction**	Identify proposals, objections, evidence, corrections, agreements, decisions and action items; associate relevant evidence with slides.	Structured meeting-event representation
9	**Decision Lineage & Interaction Analysis**	Construct decision chains and participant interaction relationships to trace how discussions lead to final decisions.	Decision lineage and interaction graph
10	**Participant Influence Model**	Develop explainable influence scoring using participation, evidence, interaction and decision-impact features; implement baselines.	Influence-analysis module and baseline results
11	**Evaluation & Integration**	Conduct component-wise evaluation, ablation study and human evaluation; integrate the complete pipeline and profile processing time.	Evaluation results and integrated prototype
12	**Documentation & Final Presentation**	Analyse results, finalize the project report, prepare demonstration and presentation.	Final report, prototype and presentation

Table 5.1: Timeline of the project

Chapter 6

Conclusion

6.1 Conclusion

The intended project seeks to design an offline multimodal meeting-intelligence system involving long meetings audio and PPT. The system would involve speaker-annotated transcription, audio-PPT alignment, meeting events extraction, decision reconstruction, and influence analysis of the meeting participants.

As opposed to traditional meeting summary tools that mostly concern with the content of the discussion, the intended design is meant to analyze how the decisions were made and the participants who had a significant influence on the decisions. Influence analysis will be conducted using decision relevant contribution as opposed to talking time only.

The intended system architecture is intended to be feasible for offline processing even in laptop computers, whereas the evaluation procedure would measure transcription, diarization, alignment, events extraction and influence analysis performances against relevant baselines.

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MULTIMODAL AI

PROJECT REPORT

"Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis"

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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1. Resource Allocation and Risk Analysis

1.1 Resource Allocation

The project will need local storage space, computing power, and various software libraries for processing the multimodal data of meetings.

- **Storage**

  The AMI Meeting Corpus contains a huge amount of audio recordings of the meetings. The raw audio recordings, consisting of headset and microphone array channels, take up more than 80 GB of storage space. There is also the need for additional storage space to store processed audio, transcripts, speaker diarization outputs, extracted presentations, and semantic embeddings.

- **Computational Resources**

  Computing power is needed primarily for the speech-processing parts of the project. Faster whisper is used for Automatic Speech Recognition (ASR). Speaker diarization is performed using the library pyannote.audio [1], [2]. Silero VAD is used for voice activity detection before the speech processing stages [3].

  As ASR and speaker diarization require much computing power, the pipeline of the project is designed to work as an offline modular workflow where the intermediate outputs can be cached and reused instead of performing the same processing on the raw recordings again. ASR also uses int8 quantization for reducing the computational load.

- **Software Resources**

  Python programming language is used as the development environment. Some of the main software libraries used in the pipeline are listed below:

  - Faster-whisper for Automatic Speech Recognition [1]
  - Pyannote.audio for speaker diarization [2]
  - Silero VAD for voice activity detection [3]
  - Python-pptx for extracting presentation content [4]
  - Sentence-transformers for generating semantic embeddings [5]
  - LibreOffice for conversion from legacy .ppt to .pptx format

  These modules are separated into stages to test the modules and reuse their output.

1.2 Risk Analysis and Mitigation

- **Data Format Incompatibility**

Presentation files from the AMI corpus consist of legacy .ppt files; however, the presentation-extraction pipeline requires the .pptx format.

Mitigation: A preliminary step is introduced, where legacy presentation files are batch-converted to .pptx format via headless LibreOffice, followed by extraction of the content.

- **Computational Bottlenecks**

Transcription and diarization of audio are computationally demanding tasks, especially for long recording meetings.

Mitigation: An independent processing pipeline with modular structure and intermediate JSON outputs, like transcripts and alignments, can be cached. This prevents rerunning computationally expensive processing steps during development and testing.

- **Acoustic Challenges**

Overlapping speech, noise, and room reverberation in the recording from a meeting room can influence the performance of Automatic Speech Recognition and speaker diarization.

Mitigation: Speech activity detection and ASR segmentation are aligned by temporal overlap with speaker diarization boundaries. It allows to match a speech segment with its appropriate speaker even if the boundaries from these two tasks do not perfectly match.

- **Multimodal Alignment**

The project consists of combining spoken dialogue and presentation slides. Misalignment in time or semantics between these two streams can impact decision making and influence analysis.

Mitigation: Timestamps for the audio track are preserved through the whole speech processing pipeline, while the content of presentations is translated into textual representations and semantic embeddings. These two types of information can be later aligned based on both temporal and semantic information.

2. Description of the Selected Dataset

This project uses the AMI (Augmented Multi-party Interaction) Meeting Corpus, which is a multimodal database dedicated to the study of meetings and human interaction. The corpus comprises about 100 hours of synchronized meeting recordings made in several instrumented meeting rooms [6].

Most of the meetings consist of the group of four people, playing different roles in the fictional design-project scenario. In addition, the database includes natural meetings, which gives a variety of types of interaction represented in the database [6], [7].

The following multimodal inputs are included in the dataset:

- Close-talking headset microphone recordings of individual participants
- Microphone array recordings of the whole room
- Video streams
- Electronic copies of presentations and documents shown in meetings
- Word-level speech transcripts
- Dialogue-act annotations
- Named-entity annotations
- Meeting summaries
- Decision-related annotations [6], [7]

Synchronization of the sources mentioned above is especially critical for the proposed project since information in one modality can be correlated with information in another one using the timeline of the meeting.

As regards the speech-processing component, word-level transcripts and dialogue-act annotations can be used as reference information for analysing the conversation behavior. Decision-related annotations are appropriate for the evaluation of the decision-analysis component of the proposed system. Presentation and document data contain additional context, which can be correlated with the topics mentioned by participants.

This database has been chosen due to the combination of speech, speaker information, presentation material and manual annotations available in it. It includes all required multimodal inputs and reference data for the evaluation of the proposed offline meeting-intelligence system [6], [7].

3. Modalities Chosen for the Project

The meeting intelligence system proposed is centered around two major modalities: spoken audio and visual/document information. Such modalities were chosen because they complement each other in terms of information available about the meeting: audio captures the interactions between meeting participants, while presentation captures the contextual information about the topic discussed.

3.1 Spoken Audio Modality

Spoken audio modality captures the process of verbal communication between meeting participants. The corpus provided by the AMI dataset includes close-talking headsets and room microphones array audio recordings [6], [7].

Processing of such audio can be decomposed into several steps:

1. Performing Voice Activity Detection (VAD) in order to detect speech regions;
2. Speaker Diarization that will tell us who was speaking at any particular moment in time;
3. Automatic Speech Recognition (ASR) in order to transcribe audio to text;
4. Speaker-attributed transcript generation to link recognized speech to particular participants;
5. Temporal alignment of ASR segments and speaker-diarization segments.

The result will contain the information about what was said, who said it, and when. Such information is essential in analysis of discussion flow, proposals, responses, decision traces, and participant influence.

The speech processing components used are Silero VAD, faster-whisper, and pyannote.audio [1]–[3].

3.2 Visual/Document Modality

The second chosen modality is the presentation and document content provided in the meetings.

The AMI corpus has the digital copy of presentations and documents used in the meetings [6], [7].

This modality is able to provide some information that might not be available from the transcript itself. The presentation content may include:

- Slide text
- Tables
- Notes

- Design alternatives
- Technical details
- Any other material related to the discussion

The first step in the process of document analysis is converting legacy presentation files into format which is suitable for programmatically analysing. Extracting relevant information from these files is done with the help of python-pptx [4]. After extraction this information may be represented by semantic embeddings created with sentence-transformers [5].

This kind of representation allows aligning the presentation content with the corresponding discussion.

3.3 Rationale for Modality Selection

The two modalities provide complementary information:

Modality	Primary Information
Spoken Audio	Who said what, when, and how the discussion progressed
Presentation/Document	What topic, design, or material was being discussed

Combination of these sources will allow the system to analyse the meeting in a way that goes beyond just speech transcription. More specifically this combination will allow performing the tasks of decision analysis and participant influence analysis.

Though the AMI corpus includes video recordings, it was decided not to use this modality for the current implementation of the project.

4. Exploratory Data Analysis (EDA) and Preliminary Data Exploration

The chosen AMI corpus dataset comprises multiple modalities: audio data, presentations, and annotations. As such, EDA was adjusted according to the specifics of each chosen modality. The initial investigation was aimed at exploring the dataset structure, audio properties, conversational annotations, decision annotations, and presentation formats prior to applying the major machine learning components.

4.1 Audio Data Exploration

The audio data was explored both on the file and metadata level to analyze the recording sources available and whether they are appropriate for speech processing. Specifically, the AMI corpus dataset includes both headset audio data and room microphone-array audio data [6], [7].

These two recording settings have distinct properties. On one hand, headset data is more related to individual participants, while room-array data captures the meeting atmosphere and can include background noise, reverberation, and overlapping speech.

In addition, the available WAV audio data recordings were analyzed in terms of their recording length and audio properties. The meeting duration analysis at the corpus level was made using the available audio recordings to assess the variability in meeting lengths. This aspect is important because the system cannot assume all meetings to have identical duration.

The initial analysis showed that audio pre-processing is required prior to ASR and speaker diarization processes. As such, the pre-processing pipeline was implemented to normalize the selected audio input data by converting multi-channel audio to mono and using 16 kHz sampling rate.

4.2 Dialogue-Act Exploration

The AMI data set also features manually annotated dialogue acts, which characterize the various forms of communicative behavior [6], [7]. The annotations were analyzed in order to reveal the linguistic structure of the meetings.

In total, 116,357 annotated dialogue act occurrences were found in the analysis of the dialogue-act XML files. The most common categories and their distributions are shown below:

Dialogue Act	**Count**	**Percentage**
Inform	33,484	28.8%
Backchannel	21,910	18.8%
Assess	21,391	18.4%
Fragment	16,796	14.4%
Suggest	9,391	8.1%
Stall	8,459	7.3%
Elicit-Inform	4,240	3.6%
Other	686	0.6%
Total	**116,357**	**100.0%**

The analysis involved the parsing of the dialogue act XML annotations and conversion of the AMI-specific dialogue-act IDs into dialogue-act categories.

The distribution indicates that the conversation during the meetings contains several types of interactions besides simple informative utterances. In particular, the Inform, Backchannel, Assess, and Fragment categories are prevalent in the annotated conversation, while Suggest and Elicit-Inform categories are related to the proposal and request for information respectively.

It is important for the current project since neither decision formation nor influence of participants on the process can be accurately inferred based on isolated keywords. The role of an utterance within the conversation and its place should be considered as well.

4.3 Decision Annotation Exploration

Decision annotations have also been discovered in the AMI corpus. It is due to the need to look at the structure of decision XML files and understand how decisions are represented in the corpus.

These decision annotations are helpful as reference data for the decision-analysis component that is proposed. The data will help to evaluate the effectiveness of the system in finding important decision points and linking them to the discussion that surrounds the decision.

Analysis suggests that decisions have to be treated in the context of preceding conversational actions. Actions like Suggest and Assess, which are found through the dialogue-act analysis, can provide context clues to follow the progression of the discussion towards the decision.

4.4 Presentation and Document Exploration

Presentation data were explored separately from audio data since they had different formats. The AMI corpus contains digital copies of presentations and documents used in meetings [6], [7].

During the exploration, it was found that there is a problem with format compatibility: presentation files may be in the legacy .ppt format, while the .pptx format is required by the extraction workflow chosen.

In order to solve this problem, document preprocessing workflow has the following steps:

1. Conversion of legacy .ppt files to .pptx using LibreOffice.
2. Textual and structural content extraction of each slide in .pptx using python-pptx [4].
3. Textual preprocessing of the extracted content.
4. Generation of semantic embeddings using sentence-transformers [5].
5. Alignment with the spoken discussion through these representations.

Thus, it was found that document-format conversion is necessary preprocessing stage and not an optional transformation.

4.5 Cross-Modal Data Exploration

Given the fact that the project involves both audio and documents, their relationship was explored during preliminary exploration as well.

The representations of interest can be described as follows:

Audio -> Speech Segments -> Speaker Information -> Transcript

Presentation -> Slide Content -> Semantic Representation

These representations can then be linked by means of temporal and semantic information. For instance, the spoken discussion of some design option can be linked to the presentation content related to the same subject matter.

This intermodal relationship is critical for the decision-lineage part of the project, since the context surrounding the particular decision can be distributed among the spoken conversation and the material itself.

4.6 EDA Findings and Preprocessing Implications

During the preliminary exploration, several observations were made which would have an impact on the pipeline design:

Observation	Implication for the Pipeline
Multiple audio recording sources	Audio standardization is required
Noise, reverberation and overlapping speech may occur	VAD and speaker-aware processing are required
Different meeting durations	Meetings need to be processed independently
Multiple dialogue-act categories	Transcript analysis should consider conversational context
Decision annotations are available	They can provide reference information for decision analysis
Presentations may use legacy `.ppt` format	Format conversion is required before extraction
Audio and document data contain complementary information	Cross-modal alignment is required

Overall, the EDA was performed as a pre-processing and modality-specific data exploration process rather than regular tabular EDA.

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MULTIMODAL AI

PROJECT REPORT

"Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis"

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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AY 2026-27

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1. Project Novelty

The common meeting summarisation tools available today use only the cloud-based generative language model. That decision presents two big issues in the corporate or legal world: the model itself can get its data wrong — it may "hallucinate" a decision or misattribute a quotation — and the voice itself must leave the premises and be stored in another company's servers, which likely isn't a desirable option for confidential boardroom discussions. The number of failures is minimized by avoiding the above two failure modes and hence, InfluenceIQ was created.

The other thing was, of course, training a custom model from scratch, and in fact ours was only about 100 hours worth in their AMI Meeting Corpus — that's not enough to train a competitive acoustic or language model. Instead we divided the problem in two parts and used the tools that were appropriate to each part: "train our own" and "hand everything to an LLM"

- Meetings: Perception (hearing the meeting): Faster-whisper (for transcription) and pyannote.audio (for diarisation), both in the inference mode. To put it into perspective: We have used only 100 hours of audio for pretraining Whisper, and already it gets transcription accuracy that our own 100 hours of audio would never have matched, let alone surpassed.
- Reasoning (understanding the meeting): a proprietary, deterministic, rule-based extraction engine instead of a generative model. Our pipeline reads the aligned discussion transcript and uses syntactic and keyword heuristics to extract the correct sentences where a proposal, objection, agreement, revision or action item took place, instead of asking an LLM to "summarise" the discussion.Because every line the system surfaces is a sentence that was literally spoken, at a literal timestamp, by a literal speaker, the output is 100% traceable back to the source audio — something a generative summary simply cannot promise. This is the core novelty we are pursuing: verifiable, non-hallucinating meeting intelligence, not another paraphrasing layer on top of an LLM.

Each line produced by the system is a true, literal sentence, spoken at a literal date and by a literal user, so there's a 100% traceability back to the source audio, which generative summaries can't offer. This is what we're working on in its essence, something that is not hallucinating but verifiable and intelligent/meaningful during the meeting, not another paraphrasing layer on top of an LLM.

The second novel idea is that 'the louder you speak, the more influence'. Everyone who attempts to determine who was responsible for the meeting defaults to the easiest to measure signal - airtime. We created an additional scoring system based on whether the contributions

of a person helped a decision, whether their objections were addressed, whether their revisions were adopted, whether their line helped to bring the line to a close, as opposed to whether they took the floor for a long or short period of time. This is more than a theoretical assertion—it is plainly reflected in the results of our own tests, as revealed in Section 3 below.

Lastly, the entire pipeline is intended to be operational on the outskirts. The whole perception layer is run on regular consumer devices (with the faster-whisper feature provided by CTranslate2) instead of the full-size models, and no audio will ever leave the machine on which it was recorded.

2. Methodology

The system is designed to be a four stage pipeline that converts a raw meeting recording (with or without a slide deck) into structured, evidence-based meeting intelligence. Each of the phases is intentionally designed to be independent, which means that if a component of it fails or has a problem, it does not silently cause another component to fail or have a problem in a later phase (for example, if the segment of the diarisation had a problem, then this does not affect the reasoning performed in a later phase).

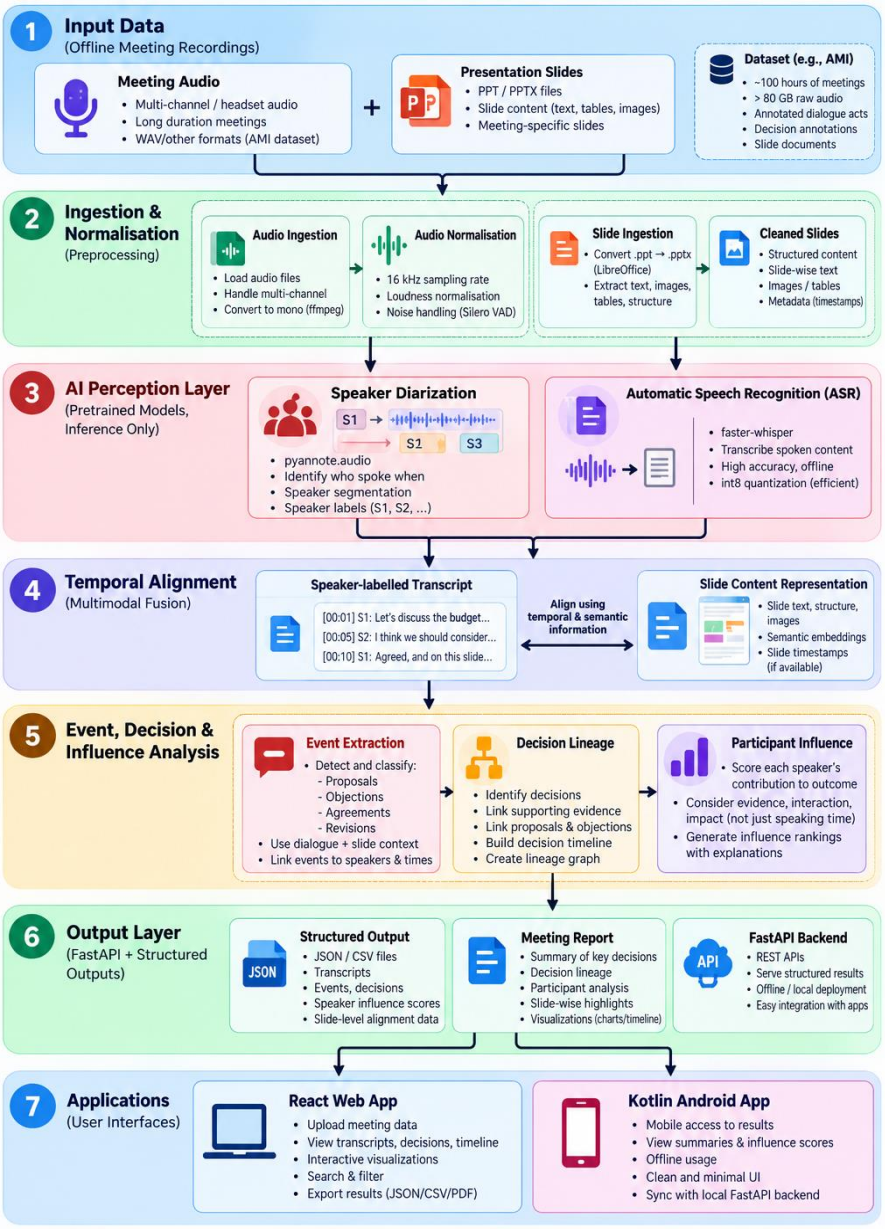


Figure 1: InfluenceIQ's four-phase offline pipeline, from raw audio to the React/Kotlin front ends.

2.1 Ingestion & Normalisation

Typical recordings are not clean — recordings on a dictaphone or phone on a boardroom table will have variable volume and whatever sample rate the device ends up using. All incoming data is processed by ffmpeg to be sure of a strict 16 kHz mono WAV format and a loudnorm filter to ensure that quieter speakers are played equally loudly to the downstream models as the loudest speaker in the room.

2.2 AI Perception Layer

The normalized sound is fed to two basic models simultaneously: pyannote.audio extracts the “who spoke when” information and faster-whisper translates the content. They are both used in inference mode only; that is, we load pre-trained weights and don't fine-tune them, hence we're able to achieve production grade accuracy with a small dataset. Once both outputs are available, we align text tokens mathematically to speaker boundaries, provided by Pyannote, to get a single speaker labelled transcript of Whisper.

2.3 Extractive NLP Engine (core contribution)

This is where the transcript gets converted into intelligence and is what we think is our intellectual contribution to the system. Three modules based on rules process the transcript sequentially, rather than passing it through an EM-based model:

- events.py — reads the transcript and tags a Proposal, Agreement, Objection and Revision as they are encountered using keyword and syntactic heuristics.
- decisions.py processes the weaving together of events, with regard to a topic; it keeps track of a topic from its first occurrence, through any objection/revision it gathers, until it reaches the point where it is confirmed as a final decision (if any).
- influence.py — this scores each player's interaction with that lineage, taking into account whether their revision was an accepted one or not, whether they said the final "accept" or not, etc., not just the number of words spoken.

2.4 Presentation Layer

The structured output – decisions, action items, participant scores and the evidence trail – is output as JSON and CSV, with a FastAPI backend to serve these outputs. The same meeting intelligence can be used in a React web app and a Kotlin Android app, which means that people can review results at their desks or right after the meeting on their phones.

3. First-Level Results

To validate the pipeline end-to-end, we ran it against meetings from the AMI corpus. The numbers below are directly from one representative test run (without any manual cleaning up), meeting_288ab746, and are what the system produced.

Metric	Value
Meeting analysed (test run)	AMI corpus — meeting_288ab746
Duration	00:35:33
Participants detected	6 (5 identified speakers + 1 unresolved segment)
Presentation slides linked	8
Raw conversational events detected	162
Confirmed decisions	1
Action items extracted	3
Tracked participant interactions	17

3.1 Decision Lineage Tracking

The system did not merely provide a summary of a topic's journey through the decision-making process to closure, but actually re-created the entire process: who initiated it, who opposed it, who edited it, or who finally agreed to close it. At this meeting, a single thread of discussion was followed through to a definite conclusion:

Decision ID	decision_001
Status	Confirmed
Topic / final text	“Either, whatever we decide the default to be…”
Participants involved	SPEAKER_04, SPEAKER_05, SPEAKER_00, SPEAKER_02, SPEAKER_01, UNKNOWN
Linked slide	slide_04

The full lineage the system reconstructed was: Then a question by SPEAKER_04 opened the topic, to which he raised an objection and SPEAKER_00 and SPEAKER_02 added objections, and there was finally a convergence on a common agreement, followed by a confirmed final decision by SPEAKER_00, who added the key revision. This is the type of chronological, evidence-based narrative that a summary would fail to keep a coherent thread on, such as here, where each step can be traced back to its speaker and the time of the speech.

3.2 Action Item Traceability

Being extractive, not generative, every action item that emerges from the pipeline is assigned to the speaker and the specific time and date that the action item was stated, nothing is assumed or paraphrased.

Action item (as spoken)	Speaker	Timestamp
"I'm just going to hopefully add something by next week that we can sort of try and…"	SPEAKER_00	00:20:16
"I'm not afraid to take that."	SPEAKER_02	00:31:59
"Like, take care of yourself."	SPEAKER_05	00:34:19

3.3 Participant Influence vs. Speaking Volume

How participants affect the speaker's voice.Participants' effects on the speaker's voice. This is the biggest finding from the first pass – this was the one we were testing our core hypothesis on – does the speaker that speaks the most drive the decision? The answer to the question is no, as indicated in the table and chart below.

Participant	Speaking Time	Speaking Rank	Influence Score	Influence Rank
SPEAKER_04	141.8 s	#1	0.3250	#2
SPEAKER_00	70.2 s	#2	0.3667	#1
SPEAKER_05	56.0 s	#3	0.0833	#5
SPEAKER_02	50.0 s	#4	0.1333	#3
SPEAKER_01	3.0 s	#6	0.1167	#4
UNKNOWN	6.0 s	#5	0.0167	#6

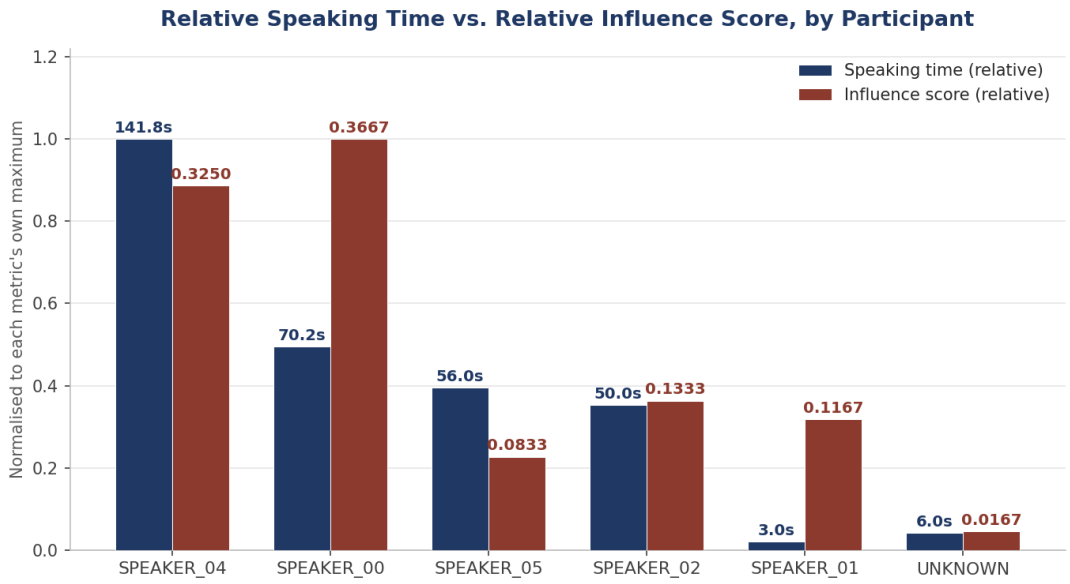


Figure 2: Relative speaking time vs. relative influence score, normalised to each metric's own maximum.

A pure acoustic analysis would have concluded that SPEAKER_04 held the meeting, as he led in both number of words spoken (141.8) and number of turns taken (66), more than double of any other speaker. We took the #2 spot in our influence engine. SPEAKER_00 speaks less (70.2 seconds), is credited with the adopted revision and the role of "final decision-maker" which makes them rank 1. The best example is SPEAKER_01, who was able to speak for just three seconds throughout two speaking turns, and yet outranked a speaker who spoke almost twice that long, SPEAKER_05, because SPEAKER_01's three seconds was directly anchored to a slide relevant to a decision.

We consider this as step one and not the end goal, as we need to see an adequate evaluation (a comparison of the system's rankings to human-annotated ground truth across multiple meetings) to be the next big step. Even at this point, this is a big enough disparity between the speaking-time ranking and the influence ranking to support the central thesis: They measure two different things, and a system that only measures the former is missing the point of the meeting.

4. Summary

It may seem like different varieties of the same theme, but the common thread throughout novelty, methodology, and this first round of results is this: create a system whose output can be verified against source recording, in other words, it's not something you have to take on faith. That's the key, and its a nice throw out to the two issues that enterprise and legal teams have had to contend with – hallucination and cloud data exposure – when considering the use of AI meeting tools – which are solved by using foundation models only for perception, while the reasoning remains strictly deterministic).

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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Artificial Intelligence and Machine Learning Department

LAB REPORT

Offline Multimodal Meeting-Intelligence System

Course: Multimodal Artificial Intelligence Lab (2026-27)

Literature Review for Project

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DETAILED SCOPUS PAPER ANALYSIS

Core Problem Statement of Project : An offline system that combines long-form meeting audio perform speaker attribution, multimodal alignment, event extraction, decision-lineage reconstruction, and explainable participant-influence analysis.

1. Executive Summary

The selected literature forms a coherent technical foundation for the proposed meeting-intelligence system. The papers mainly address five enabling problems: low-latency speaker spotting/diarization, streaming speaker-attributed ASR, speech enhancement and separation, audio-visual diarization/concurrent-speaker detection, and recognition of overlapping speech.

Across the literature, the most important lesson is that long-form meeting analysis should not rely on a single monolithic model. The strongest engineering direction is a modular but tightly aligned pipeline in which preprocessing, speaker tracking, transcription, multimodal fusion, event extraction, and decision analysis exchange timestamps, confidence values, and speaker representations.

For the proposed project, the literature also exposes a clear research gap: these works largely stop at ‘who spoke when’ or ‘who spoke what when’. They do not provide a complete mechanism for connecting a speaker’s contribution to presentation evidence, tracing how that contribution changes the discussion, reconstructing decision lineage, and computing an explainable influence score.

Therefore, the literature supports positioning the project not as another meeting transcription system, but as a decision- and influence-analysis layer built on top of robust multimodal speech processing.

2. Main Findings at a Glance

Technical area	Key lesson	Best-matching paper(s)	Project use	Remaining gap
Low-latency speaker detection	Optimize diarization for the downstream task, not only DER.	Patino et al. (2018)	Fast speaker initialization and target detection	No semantic/decision reasoning
Streaming SA-ASR	Speaker identity can be attached at token level during streaming recognition.	Kanda et al. (2022)	Speaker-content-time representation	No PPT or influence
Speech enhancement	Small causal models can make local processing practical.	Liang et al. (2023); Lay et al. (2025)	Offline/on-device preprocessing	Enhancement is not identity/causality
Continuous separation	Long sequences and overlap require special architectures.	Li et al. (2022)	Overlap front-end	Separated streams still need attribution
Fast/distilled separation	Teacher-student distillation lets a small model approach a large model's accuracy on real meeting-room audio.	Chen et al. (2021)	Laptop-deployable overlap front-end	No speaker-to-decision linkage
Audio-visual diarization	Visual evidence improves robustness when audio is ambiguous.	Wang et al. (2023); Ishizuka et al. (2009); Eliav & Gannot (2026)	Speaker association and overlap handling	No presentation evidence
Online correction	Streaming diarization benefits from bounded backtracking.	Zhen et al. (2024)	Correct early speaker errors before final scoring	No decision-level correction
Overlap detection/counting	Spatial features and speaker count improve overlap handling.	Cornell et al. (2022)	Gate downstream models and estimate uncertainty	No semantic contribution model
End-to-end multi-talker ASR	Joint modeling can reduce the cost of strict separation→ASR cascades.	Tripathi et al. (2020)	Alternative ASR architecture	Limited real-meeting generalization

3. Overall Literature Synthesis

- The literature strongly validates the importance of low latency, overlap handling, and speaker-attributed transcription for meeting systems.
- Audio-only pipelines remain vulnerable to noise, reverberation, simultaneous speech, short turns, and similar voices.
- Audio-visual fusion provides complementary evidence, especially for active/concurrent speaker detection and speaker-to-person association.
- Streaming systems need mechanisms for maintaining speaker identity over long recordings and correcting early mistakes.
- Teacher-student distillation (Chen et al., 2021) shows that laptop-class hardware can run fast separation models without a large accuracy penalty, directly supporting the offline/local-processing requirement.
- The selected papers do not directly solve presentation-aware evidence grounding or participant influence on decisions.
- The proposed research opportunity is therefore at the integration and reasoning layer: speaker-attributed multimodal events → evidence links → interaction graph → decision lineage → explainable influence.

4. Detailed Analysis of Each Paper

1. Enhanced low-latency speaker spotting using selective cluster enrichment

Authors: Patino, J.; Delgado, H.; Evans, N.

Research problem: Low-latency speaker spotting (rapid detection of a known target speaker) in multi-speaker audio streams.

Core method: Reframes speaker diarization around the downstream speaker-spotting task rather than around minimising the diarization error rate (DER), and introduces a selective cluster enrichment algorithm that uses the target-speaker model to guide which segments the diarization sub-system models, improving cluster purity for the target speaker.

Dataset / setting: Built on the AMI Meeting Corpus, using LLSS-specific protocols (not the standard AMI diarization protocols) designed and released by the authors to support low-latency speaker-spotting evaluation and reproducibility.

Main results: The proposed LLSS system detects target speakers with a 16% equal error rate (EER) after the target speaker has been active for only 15 seconds within the multi-speaker stream.

Strengths: Strong application-oriented design; explicitly optimises the upstream diarization component for the final detection task; demonstrates a useful latency/accuracy trade-off; uses a genuine multi-party meeting corpus (AMI).

Limitations: The target is known-speaker spotting rather than full unconstrained meeting diarization; results are reported on one benchmark family (AMI-derived) and may not generalise to arbitrary institutional meeting rooms or speaker counts.

Direct relevance to the proposed project: Useful for fast speaker initialisation, target-speaker detection, and the broader design principle that intermediate components (diarization) should be optimised for the final meeting-intelligence objective rather than DER alone.

Research gap exposed: Does not model semantic content, PPT evidence, decision lineage, or participant influence.

Critical interpretation

This paper is important because it changes the optimisation target: a diarization sub-system should be designed around the downstream application. For the project, this supports optimising speaker tracking not merely for a low DER, but for reliable speaker-event attribution and later influence analysis.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

2. Streaming Speaker-Attributed ASR with Token-Level Speaker Embeddings

Authors: Kanda, N.; Wu, J.; Wu, Y.; Xiao, X.; Meng, Z.; Wang, X.; Gaur, Y.; Chen, Z.; Li, J.; Yoshioka, T.

Research problem: Streaming joint speech recognition and speaker attribution under overlapping speech.

Core method: Uses token-level Serialized Output Training (t-SOT) together with an encoder-decoder-based speaker embedding extractor. A 't-vector' speaker representation is estimated synchronously for every recognised token, enabling joint speaker identification/diarization and multi-talker transcription in a single streaming pass.

Dataset / setting: LibriSpeechMix and LibriCSS.

Main results: The authors report substantially better accuracy than a prior streaming speaker-attributed ASR baseline, and comparable or in some conditions better results than a state-of-the-art offline speaker-attributed ASR system, while operating in a streaming (low-latency) mode.

Strengths: Very relevant to long-form, low-latency speaker-attributed transcription; token-level speaker representations directly connect individual words with speaker identity, avoiding a hard diarization/ASR boundary.

Limitations: Simulated and benchmark datasets (LibriSpeechMix, LibriCSS) do not fully reproduce confidential institutional meetings; semantic/decision-level analysis is outside the model's scope.

Direct relevance to the proposed project: Provides a strong alternative to a strict diarization-then-ASR cascade. Token-level speaker attribution can give cleaner speaker-content-time units for downstream event and influence analysis.

Research gap exposed: No PPT alignment, evidence verification, decision causality, or influence scoring.

Critical interpretation

This is one of the closest technical matches to the project's audio backbone. Token-level speaker embeddings create a fine-grained unit containing speaker, lexical content, and time — ideal for later event and interaction graphs.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

3. Real-Time Diffusion Buffer for Speech Enhancement On A Laptop

Authors: Lay, B.; Makarov, R.; Gerkmann, T.

Research problem: Live, real-time demonstration of diffusion-based single-microphone speech enhancement that removes background noise in reverberant, noisy rooms (e.g. loud babble) on ordinary laptop hardware.

Core method: Packages the authors' 'Diffusion Buffer' framework — a sliding-window/temporal diffusion process that progressively denoises audio frames as they age inside a buffer, assigning more noise to frames closer to the present — as an interactive demo. Conference attendees can toggle the enhancement on/off and adjust the quality/latency trade-off live, experiencing sub-second latency directly.

Dataset / setting: No benchmark corpus is evaluated in this 2-page demo paper itself; it is a live show-and-tell running on conference-room audio (loud babble noise, reverberant single-microphone capture). Formal quantitative evaluation (PESQ/WVMOS/latency numbers) belongs to the companion full paper, 'Diffusion Buffer: Online Diffusion-based Speech Enhancement with Sub-Second Latency' (Lay, Makarov, Gerkmann, INTERSPEECH 2025), which reports algorithmic latencies of roughly 320–960 ms on a laptop-class NVIDIA RTX 4080 GPU with real-time factor below 1.

Main results: As a demonstration paper it reports no independent WER/PESQ table; the underlying Diffusion Buffer method it showcases achieves comparable quality to offline diffusion enhancement at 320–960 ms latency, with a controllable quality/latency dial (later Streaming/Buffer follow-up work pushes this toward tens of milliseconds).

Strengths: Shows that generative (diffusion-based) enhancement can be made interactive and run live on consumer laptop hardware rather than only in offline batch evaluation; makes the latency/quality trade-off tangible.

Limitations: Being a 2-page demonstration paper, it does not itself provide a controlled dataset, ablation, or statistical evaluation; those numbers must be sourced from the companion full Diffusion Buffer paper, and should not be attributed to this demo paper directly.

Direct relevance to the proposed project: Relevant whenever meeting recordings contain noise or reverberation. A lightweight, local, laptop-class enhancement front-end (as demonstrated here) could improve diarization and ASR quality while preserving the offline/privacy requirement of the proposed system.

Research gap exposed: Does not address speaker attribution, multimodal alignment, or influence.

Critical interpretation

The paper's main value is as an implementation and feasibility lesson: generative enhancement can be adapted to run interactively, in real time, on a laptop. For the project, enhancement should be treated as an optional, confidence-aware front-end rather than as the main research contribution — and any numerical claims should be drawn from the companion full paper, not this demo abstract.

Dimension	Assessment	Project implication
Technical maturity	Demonstrated feasible on consumer hardware; formal numbers require the companion paper	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

4. Lite-RTSE: Exploring a Cost-Effective Lite DNN Model for Real-Time Speech Enhancement in RTC Scenarios

Authors: Liang, X.; Zhang, L.; Wu, Z.; Xu, R.

Research problem: Efficient real-time speech enhancement for computationally constrained real-time-communication (RTC) hardware.

Core method: Combines low-dimensional LSTM units with a multi-order convolution block and a two-stage complex-spectrum reconstruction scheme (masking followed by residual reconstruction).

Dataset / setting: Not fully specified in the supplied abstract-level record.

Main results: The model contains only 1.56 million parameters and requires 0.55 GMAC/s while achieving denoising performance competitive with state-of-the-art enhancement systems.

Strengths: Excellent engineering fit for laptop/edge deployment; explicitly balances quality, model size, and computation for real-time use.

Limitations: Monaural speech enhancement does not by itself solve overlapping speakers, and may suppress useful speaker information if run too aggressively ahead of diarization.

Direct relevance to the proposed project: Good candidate for a lightweight local preprocessing module before VAD/diarization/ASR, especially since the final system must work without cloud GPUs.

Research gap exposed: No semantic or multimodal reasoning; no evidence or decision modeling.

Critical interpretation

The 1.56M-parameter, 0.55 GMAC/s footprint makes this work especially relevant to the requirement for practical local processing. A similar design philosophy can keep the meeting pipeline usable on ordinary laptops.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

5. SKIM: Skipping Memory LSTM for Low-Latency Real-Time Continuous Speech Separation

Authors: Li, C.; Yang, L.; Wang, W.; Qian, Y.

Research problem: Very-low-latency continuous speech separation for meeting audio.

Core method: Uses a time-domain separator with Graph-PIT permutation-invariant training and introduces SKIM (Skipping Memory LSTM) for efficient long-sequence modelling, targeting the long feature sequences produced by low-latency encoders.

Dataset / setting: Simulated meeting-style evaluation; the abstract identifies meeting-style audio but not a full dataset specification.

Main results: SKIM achieves performance on par with or better than DPRNN while reducing computational cost by 75%. The fastest real-time configuration reports a 17.1 dB SDR improvement with under 1 ms of algorithmic latency in the simulated evaluation.

Strengths: Highly relevant to continuous meeting preprocessing; addresses the computational burden of long sequences while preserving low latency.

Limitations: Separation can introduce artefacts and is not equivalent to speaker-identity tracking; simulated meetings may differ from real institutional recordings.

Direct relevance to the proposed project: Potential front-end for overlap handling when ASR/diarization accuracy drops during simultaneous speech.

Research gap exposed: No direct mapping from separated streams to semantic contributions, slides, or decisions.

Critical interpretation

SKIM demonstrates that very low latency does not necessarily require abandoning long-context modelling. This matters because a 1–2 hour meeting contains long-range speaker and topic dependencies even when the acoustic model itself operates on short windows.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

6. The Multimodal Information Based Speech Processing (MISP) 2022 Challenge: Audio-Visual Diarization and Recognition

Authors: Wang, Z.; Wu, S.; Chen, H.; He, M.; Du, J.; Lee, C.-H.; Chen, J.; Watanabe, S.; Siniscalchi, S. M.; Scharenborg, O.; et al.

Research problem: Benchmarking audio-visual speaker diarization and recognition in realistic far-field scenarios.

Core method: Defines Audio-Visual Speaker Diarization (AVSD) and Audio-Visual Diarization+Recognition (AVDR) tasks, combining audio and video to address both 'who spoke when' and 'who spoke what when'. The challenge establishes datasets, tracks, baselines, and evaluation conditions.

Dataset / setting: Chinese far-field, home-TV recording scenarios with 2–6 participants and TV background noise.

Main results: The challenge baselines demonstrate the feasibility of AVDR, while the accompanying analysis identifies difficult conditions including far-field video quality, TV noise, and visually indistinguishable speakers.

Strengths: Strong benchmark orientation; explicitly tests multimodal robustness in realistic noisy environments; establishes 'who spoke what when' as a concrete, evaluable task.

Limitations: The scenario is home-TV rather than PPT-driven professional meetings; it does not evaluate decision influence.

Direct relevance to the proposed project: Supports using visual/audio multimodality to resolve speaker ambiguity and overlap. Its 'who spoke what when' framing is directly relevant to the project's timestamped event representation.

Research gap exposed: The modality missing from the challenge, relative to this project, is presentation evidence and decision semantics.

Critical interpretation

MISP is particularly important because it formalises 'who spoke what when' as a multimodal task. The proposed system extends this idea by asking an additional question: what did that contribution do to the decision process?

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

7. Audio-Visual Approach for Multimodal Concurrent Speaker Detection

Authors: Eliav, A.; Gannot, S.

Research problem: Audio-visual detection of multiple simultaneously active speakers.

Core method: Early fusion of audio and visual features through cross-modal attention with a learnable [CLS] token, evaluated as a multimodal concurrent speaker detection (CSD) system.

Dataset / setting: AMI and EasyCom real-world datasets.

Main results: Experiments support the effectiveness of multimodal fusion; the work reports CSD results on EasyCom, a difficult real-world egocentric-conversation benchmark.

Strengths: Directly addresses overlap, a major failure mode for diarization and ASR; multimodal fusion can exploit visual evidence when acoustics alone are ambiguous.

Limitations: Concurrent speaker detection identifies how many/which candidates are active, not what was said or whether a contribution changed a decision.

Direct relevance to the proposed project: Useful as an overlap detector positioned before speaker attribution and event extraction; visual cues can improve reliability during simultaneous speech.

Research gap exposed: Needs integration with speaker identity, semantic event extraction, PPT evidence, and decision lineage.

Critical interpretation

Concurrent-speaker detection should be conceptually separated from speaker identity. Detecting overlap first can tell the system when speaker attribution is uncertain and when it should invoke stronger multimodal or separation models.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

8. A speaker diarization method based on the probabilistic fusion of audio-visual location information

Authors: Ishizuka, K.; Araki, S.; Otsuka, K.; Fujimoto, M.

Research problem: Online audio-visual speaker diarization using physical location evidence.

Core method: Combines speech-presence probabilities from a compact microphone array with face locations from two cameras using probabilistic fusion and online classification.

Dataset / setting: Real casual multi-party conversations with substantial overlap and short speech segments.

Main results: The authors report advantages over conventional (audio-only) diarization approaches, especially under overlap and short-turn conditions.

Strengths: Early demonstration that visual location can stabilise online speaker assignment; explicitly designed for online processing rather than full-recording hindsight.

Limitations: Older feature/model technology; location-based identity is weaker when people are occluded, off-camera, or move substantially.

Direct relevance to the proposed project: Provides historical foundation for audio-visual speaker-to-person association and supports using physical/visual context as an additional confidence signal.

Research gap exposed: No semantic contribution modeling or presentation/decision evidence.

Critical interpretation

This paper provides an early probabilistic foundation for combining physical visual location with audio evidence. Its historical importance is that multimodal speaker assignment is not new; the project must therefore focus its novelty on higher-level evidence and decision reasoning.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

9. Winner Takes It All: An Efficient Overlap-Aware Hybrid Online Diarization with Partial Backtracking Mechanism

Authors: Zhen, R.; Zhang, X.; Min, C.; Li, B.

Research problem: Efficient overlap-aware online diarization with limited correction of early streaming errors.

Core method: Combines incremental clustering and local diarization, adds a transformer segmentation model, and introduces partial backtracking to repair temporary early errors without fully revising all past outputs.

Dataset / setting: AMI, VoxConverse (VOX), and DIHARD III.

Main results: Relative DER decreases of 13.2%, 15.5%, and 9.7% are reported on the three evaluation settings respectively, with over 30% reduction in inference time; a two-pass clustering strategy is shown to further improve offline results.

Strengths: Very strong match for long meetings: streaming systems inevitably make early mistakes, and partial backtracking offers a practical compromise between stability and correction.

Limitations: Backtracking increases system complexity; diarization correction still does not guarantee correct semantic or causal interpretation.

Direct relevance to the proposed project: Highly relevant to long-form meeting processing. The project should consider bounded backtracking so later evidence can correct speaker assignments before influence scores are finalised.

Research gap exposed: No multimodal presentation evidence or decision-influence model.

Critical interpretation

Partial backtracking is highly relevant to the project's explainability requirement. If later evidence reveals that an early speaker label was wrong, the system should be able to revise dependent event and influence records rather than freezing an early error forever.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

10. Overlapped Speech Detection and speaker counting using distant microphone arrays

Authors: Cornell, S.; Omologo, M.; Squartini, S.; Vincent, E.

Research problem: Unified overlapped speech detection (OSD) and speaker counting for distant microphones.

Core method: Frames VAD, OSD, joint VAD+OSD, and speaker counting as related supervised classification problems. Compares TCN and Transformer architectures and explores early/late fusion of acoustic and spatial microphone-array features.

Dataset / setting: AMI, CHiME-6, and a purpose-built multichannel synthetic dataset.

Main results: The Transformer architecture performs best among those tested; neural spatial-localisation features improve over single-channel features; training on speaker counting improves OSD compared with VAD+OSD training alone.

Strengths: Strong real-world relevance for meeting audio; demonstrates that spatial information and task formulation both matter.

Limitations: Speaker counting is not identity attribution; distant-microphone conditions require careful hardware assumptions.

Direct relevance to the proposed project: Can be used as an overlap/counting gate before diarization and ASR, reducing downstream confusion and helping estimate uncertainty.

Research gap exposed: No visual modality, presentation evidence, or semantic/decision reasoning.

Critical interpretation

This paper supports an explicit overlap/counting stage. Overlap is not just an ASR problem: it changes diarization confidence and should therefore propagate uncertainty into later influence scores.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

11. Ultra fast Speech Separation Model with Teacher Student Learning

Authors: Chen, S.; Wu, Y.; Chen, Z.; Wu, J.; Yoshioka, T.; Liu, S.; Li, J.; Yu, X.

Research problem: Making Transformer-based continuous speech separation fast enough for practical deployment on edge devices without losing the accuracy of large Transformer/Conformer separators.

Core method: Trains a small 'student' Transformer separator (fewer encoder layers) using layer-wise Teacher–Student (T-S) learning against a larger Conformer 'teacher' model, combined with an objective-shifting mechanism so the student learns from intermediate teacher representations as well as the final target. Large amounts of unlabeled speech are exploited during distillation.

Dataset / setting: Training: roughly 2,000 hours of LibriVox-derived mixtures plus 600 hours of in-house (noisier, no clean reference) mixtures, created by mixing single-speaker utterances. Evaluation: LibriCSS — 10 hours of concatenated and mixed LibriSpeech utterances played back and re-recorded in a real meeting room — tested in both single-channel and seven-channel settings, with both utterance-wise and continuous-input evaluation protocols.

Main results: Compared with a same-size Transformer-small model trained from scratch, the proposed T-S learning method reduces word error rate (WER) by more than 5% relative for both single-channel and multi-channel separation on LibriCSS. Exploiting additional unlabeled data for distillation yields a further 10%+ relative WER improvement, allowing the small/fast student model to approach the larger Conformer teacher's accuracy at a fraction of its inference cost.

Strengths: Directly targets the deployment gap between large/slow separators and small/fast ones; evaluated on a realistic meeting-room dataset (LibriCSS) rather than pure simulation; gives a clean latency/accuracy trade-off story achieved through model-size reduction plus distillation rather than only architectural tricks.

Limitations: Still a separation-only front-end that must feed a downstream ASR system; LibriCSS, although recorded in a real room, replays concatenated/simulated LibriSpeech content rather than genuinely spontaneous multi-party meeting speech; no semantic, evidence, or decision-level reasoning is attempted.

Direct relevance to the proposed project: Confirms that fast, laptop-deployable separation front-ends are achievable via teacher-student distillation without major accuracy loss — directly supporting the project's local/offline-processing requirement, and a good candidate to front the diarization+ASR stage specifically for overlap-heavy segments.

Research gap exposed: No speaker-content-to-decision linkage, no PPT alignment, and no influence modelling.

Critical interpretation

With full-text details now confirmed (via the ISCA Archive and the authors' arXiv preprint), this paper is one of the more directly transferable engineering results for a laptop-based, offline separation front-end: the real meeting-room LibriCSS benchmark and the 10%+ relative WER gain from distillation give a concrete, load-bearing data point for the project's overlap-handling layer.

Dimension	Assessment	Project implication
Technical maturity	Strong / established — verified via ISCA Archive	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

12. End-To-End Multi-Talker Overlapping Speech Recognition

Authors: Tripathi, A.; Lu, H.; Sak, H.

Research problem: End-to-end recognition of overlapping multi-speaker speech.

Core method: Augments an RNN-Transducer (RNN-T) with a masking model for separation of encoded audio features, multiple label encoders, a masking L2 loss, and a speaker-embedding loss to facilitate speaker tracking.

Dataset / setting: Simulated overlapping speech for training and real-life telephone conversations for evaluation.

Main results: The proposed system reports 32% WER on words in overlapping speech segments; a manual human-transcription comparison on the same test set reports 37% WER, illustrating the intrinsic difficulty of the task.

Strengths: Demonstrates that multi-talker recognition can be learned jointly rather than requiring a strict separation-then-ASR pipeline; includes explicit speaker-tracking supervision.

Limitations: Telephone speech differs from distant meeting audio; the reported WER is high and overlap remains fundamentally difficult.

Direct relevance to the proposed project: Supports exploring joint multi-talker ASR or speaker-attributed ASR instead of assuming perfect separation first.

Research gap exposed: Does not address multimodal evidence, slide alignment, decisions, or influence.

Critical interpretation

This paper demonstrates that multi-talker recognition can be modelled jointly with speaker tracking. For the project, it supports evaluating both cascaded and joint architectures rather than assuming that one pipeline is universally superior.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

5. Comparative Analysis of All Papers

Paper	Main task	Streaming	Overlap	Audio-visual	Speaker-content link	Direct project value
1	Speaker spotting	Yes	Indirect	No	Partial	High
2	SA-ASR	Yes	Yes	No	Strong / token-level	Very High
3	Speech enhancement (demo)	Yes	Not primary	No	No	Medium
4	Speech enhancement	Yes	Not primary	No	No	High for edge deployment
5	Continuous separation	Yes	Yes	No	No	High
6	AV diarization/recognition	Challenge-oriented	Yes	Yes	Strong	Very High
7	Concurrent speaker detection	Yes / real-time oriented	Core	Yes	No	High
8	AV diarization	Online	Yes	Yes	Partial	High
9	Online diarization	Yes	Core	No	Partial	Very High
10	OSD + counting	Yes / low-latency relevant	Core	Spatial audio	No	High
11	Fast/distilled separation	Yes	Core	Not primary	No (speaker-agnostic separation)	Medium–High
12	End-to-end MTASR	Streaming-capable	Core	No	Speaker tracking	High

6. Research Gap and Proposed Novelty

Capability	Covered by selected literature?	Project extension	Research importance
Who spoke when?	Yes, strongly	Use robust diarization/SA-ASR foundation	Necessary but not novel by itself
Who spoke what when?	Yes	Use token/segment speaker attribution	Core input representation
Who was speaking simultaneously?	Yes	Use OSD/CSD + multimodal cues	Important uncertainty signal
What slide/evidence was being discussed?	Not directly	Timestamp + semantic alignment with PPT	Key multimodal extension
Did a spoken claim support/contradict PPT evidence?	Not directly	Claim-evidence matching and contradiction detection	Strong project-specific contribution
How did contributions change the discussion?	Not directly	Interaction/event graph	Core reasoning layer
Which events led to the final decision?	Not directly	Decision lineage graph	Core decision-intelligence contribution
Who influenced the decision?	Not directly in this set	Influence propagation + human validation	Main research question
Can influence be explained?	Not directly	Evidence-backed score with timestamps and source events	Critical for trust and academic evaluation

7. Recommended System Architecture Based on the Literature

Layer	Recommended design	Literature / rationale	Implementation note
Layer 1 — Audio preprocessing	VAD + optional lightweight enhancement	Liang et al.; Lay et al.	Keep enhancement optional and confidence-aware.
Layer 2 — Overlap detection	OSD/CSD + speaker counting	Cornell et al.; Eliav & Gannot	Flag difficult regions and control downstream processing.
Layer 2b — Fast separation (optional)	Distilled/teacher-student separator for overlap-heavy segments	Li et al. (SkiM); Chen et al. (T-S learning)	Reserve for segments the overlap gate flags as difficult; keep laptop-deployable.
Layer 3 — Speaker attribution	Online diarization / SA-ASR / embeddings	Patino et al.; Kanda et al.; Zhen et al.	Prefer timestamped speaker-content units.
Layer 4 — Multimodal association	Audio + visual evidence where video exists	MISP; Ishizuka et al.; Eliav & Gannot	Use visual cues as complementary evidence, not mandatory input.
Layer 5 — PPT understanding	Text/table/OCR/vision extraction	Project-specific	Create slide evidence objects with slide IDs and embeddings.
Layer 6 — Cross-modal alignment	Timestamp + semantic similarity + explicit slide references	Project-specific	Align utterances to candidate slides with confidence.
Layer 7 — Event extraction	Proposal, objection, evidence, correction, agreement, decision, action	Project-specific	Convert transcript into structured events.
Layer 8 — Interaction graph	Speaker-to-speaker response/objection/support edges	Project-specific	Model discussion dynamics.
Layer 9 — Decision lineage	Contribution → evidence → modification → decision	Project-specific	Trace how decisions formed.
Layer 10 — Influence model	Participation + interaction + evidence + decision impact + temporal propagation	Project-specific	Validate against human annotations; avoid equating talk time with influence.

8. Proposed Research Methodology

A defensible experimental methodology should compare the proposed multimodal influence model against increasingly strong baselines.

Baseline	Definition	Purpose
B1 — Talk time	Influence = speaking duration	Tests whether the proposed method beats a naive participation metric.
B2 — Participation	Talk time + number of turns	Adds basic conversational activity.
B3 — Interaction	Participation + responses/objections/agreements	Captures network-level interaction.
B4 — Audio-only semantic	Speaker attribution + event extraction	Tests the value of presentation evidence.
B5 — Audio + PPT	Speaker events + slide alignment + evidence links	Tests multimodal evidence.
B6 — Proposed	Audio + PPT + interaction graph + decision lineage + temporal influence propagation	Full research model.

9. Evaluation Plan

Component	Metrics	Why
ASR	WER / CER	Measures transcription quality.
Diarization	DER / JER / speaker-attributed error	Measures who-spoke-when accuracy.
Overlap	OSD/CSD precision, recall, F1	Measures simultaneous speech detection.
Event extraction	Precision, recall, F1	Measures proposal/objection/evidence/decision/action extraction.
PPT alignment	Precision, recall, F1 / retrieval@K	Measures speech-to-slide evidence association.
Decision extraction	Precision, recall, F1	Measures final decision identification.
Influence ranking	Spearman, Kendall tau, NDCG, Precision@K	Compares model ranking with human judgments.
Explainability	Human rating of evidence sufficiency	Tests whether score explanations are understandable and traceable.
Efficiency	RTF, memory, latency, CPU/GPU utilization	Tests offline laptop feasibility.

10. Human Annotation Protocol for Influence

Influence cannot be validated using WER or DER alone. A separate human-annotation layer is required. Annotators should examine each decision and identify which contributions materially affected its trajectory.

- For every final decision, annotate the initial proposal, objections, supporting evidence, corrections/modifications, final acceptance, and action assignment.
- Assign each relevant contribution an influence category such as negligible, supporting, substantial, or critical.
- Record whether the contribution introduced new information, changed another speaker's position, resolved an objection, or directly affected the final decision.
- Use at least two independent annotators and calculate inter-annotator agreement before using the labels as ground truth.
- Treat model-derived counterfactual influence as an analytical estimate, not proof of causal influence.

11. Key Design Principles Derived from the Literature

- Optimize intermediate components for the final application, not only for generic benchmark metrics.
- Preserve timestamps throughout the pipeline; time is the common coordinate system linking audio, speakers, slides, events, and decisions.
- Represent uncertainty explicitly, especially around overlap, low SNR, missing faces, and early streaming diarization errors.
- Use bounded backtracking so later information can repair early speaker assignments without recomputing the entire meeting.
- Prefer lightweight/causal models for local deployment (including distilled separators, per Chen et al.), and reserve expensive models for difficult segments.
- Do not treat speaking time as influence; use interaction, evidence, and decision impact.
- Every high-level score should have a traceable evidence path back to transcript segments and, when relevant, slide content.
- Evaluate cascaded and joint architectures experimentally rather than assuming one is always superior.

12. Limitations of This Literature Review

- Official Scopus record links for all 12 papers, exactly as supplied for the assignment, are consolidated in Section 15; these should still be opened and cross-checked against the institution's Scopus subscription immediately before formal submission, since link tokens can be session- or account-specific.
- Most available evidence here is based on abstracts, official conference/journal archives (IEEE Xplore, ISCA Archive, ACM DL), and author preprints rather than the complete typeset Scopus abstract-and-citation record.
- Paper 41 (Chen et al., 2021) previously lacked a matching bibliographic record; it has now been fully verified against the ISCA Archive proceedings entry and the authors' arXiv preprint (arXiv:2204.12777), and the analysis in Section 4 has been completed accordingly.
- Paper 24 (Lay et al., 2025) is a 2-page INTERSPEECH Show & Tell (live demonstration) paper rather than a full experimental study; its own record carries no independent quantitative table, so numerical latency figures are explicitly sourced from the companion full paper and clearly labelled as such rather than attributed to the demo paper.
- The Scopus publication-ID string supplied for Paper 9 (Patino et al., 2018) contained characters that render ambiguously in the source list; the ID has been reproduced as supplied, but the DOI (10.23919/BIOSIG.2018.8553619) and IEEE Xplore record should be used as the authoritative cross-check.
- Several benchmark datasets used across these papers are simulated or domain-specific (e.g. LibriCSS, LibriSpeechMix, home-TV MISP recordings), so their reported numbers should not be assumed to transfer directly to university or enterprise meetings.

13. Conclusion

The literature strongly supports the technical feasibility of the project's lower layers: real-time/low-latency enhancement, overlap detection, continuous and distilled separation, online diarization, speaker-attributed ASR, and audio-visual speaker association are all active and technically mature research areas.

The stronger research opportunity is the layer above these components. The selected papers largely stop at identifying speakers, transcribing their content, or improving robustness under overlap. The proposed system can extend this foundation by aligning speaker-attributed speech with PPT evidence, extracting structured meeting events, constructing interaction and decision graphs, and estimating participant influence using evidence-backed temporal propagation.

Therefore, the recommended thesis/project contribution is: an offline, explainable, multimodal meeting-intelligence framework that goes beyond transcription to reconstruct how evidence and participant contributions propagate into decisions. The influence score should be validated against human judgments and presented as an evidence-grounded analytical measure rather than an absolute causal truth.

14. References

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